

Drosophila central complex: identifiable computation, non-identifiable implementation

Abstract

Theoretical work on emergent structure in recurrent neural networks almost never has a ground-truth comparison: discovered modules are evaluated against themselves. The *Drosophila* central complex (CX) is the rare exception — a complete connectome (Hulse et al., 2021) pairs with a known biological function (head-direction integration) and a published RNN model (Vafidis et al., 2022). We use this pairing to ask what optimisation of the path-integration task identifies when the connectome topology is given. Four constrained models are trained on the same input–output map: a Known-ODE RNN, a frozen-connectome control, a fully connected RNN, and a message-passing GNN with learned neuronal signalling functions. The frozen-connectome control fails, whereas the three trainable models recover near-perfect heading accuracy. Among them, the Known-ODE RNN is the biologically aligned recovery: it preserves the canonical CX allocation of head-direction information to EPG / EPGt and $\Delta 7$, with ER6 remaining tonic. The GNN reaches comparable behavioural performance but implements an alternative circuit, redistributing head-direction information across cell types and learning drift/message functions. Thus optimising the path-integration task recovers the ring-attractor computation, but not a unique circuit implementation. Resolving these degeneracies requires a new methodology: looking for consistency between connectome, neural activity and behavioral task.

1 Introduction

2 Methods

Recurrent neural network. The RNN model is a sign-locked discrete-time recurrent network whose connectivity is constrained to the *Drosophila* hemibrain central-complex connectome. For each neuron $i \in \{1, \dots, N\}$ the subthreshold activity $\hat{h}_i(t)$ evolves according to

$$\tau \frac{d\hat{h}_i(t)}{dt} = -\hat{h}_i(t) + \sum_j \hat{W}_{ij}^{\text{rec}} \sigma(\hat{h}_j(t)) + \sum_l \hat{W}_{il}^{\text{in}} u_l(t) + \hat{b}_i, \quad (1)$$

$$\hat{y}(t) = (\cos \hat{\theta}(t), \sin \hat{\theta}(t)) = \hat{W}^{\text{out}} \sigma(\hat{h}(t)) + \hat{b}^{\text{out}}, \quad (2)$$

where σ is the sigmoid, $\tau = 0.1$ s, $\Delta t = 0.01$ s, and the input $u(t) \in \mathbb{R}^3 = (\omega(t), \cos \theta_0 \delta_{t,0}, \sin \theta_0 \delta_{t,0})$ encodes angular velocity and a one-frame initial-heading impulse. The recurrent matrix is sign-locked to the connectome template $W^{\text{con}} \in \mathbb{R}^{N \times N}$: the trainable per-edge parameter is the magnitude $\hat{S} \in \mathbb{R}^{N \times N}$, and

$$\hat{W}_{ij}^{\text{rec}} = |\hat{S}_{ij}| \text{sign}(W_{ij}^{\text{con}}) (1 - \delta_{ij}). \quad (3)$$

The angular-velocity input is anatomically restricted to the PEN subpopulations through a strict four-scalar gate on the angular-velocity column of $\hat{W}^{\text{in}}$,

$$\hat{W}_{i,\omega}^{\text{in}} = v_s \quad \text{if } i \in s, \quad s \in \{\text{PENa}_L, \text{PENa}_R, \text{PENb}_L, \text{PENb}_R\}, \quad (4)$$

with four learnable scalars $\{v_{\text{PENa}_L}, v_{\text{PENa}_R}, v_{\text{PENb}_L}, v_{\text{PENb}_R}\}$ initialised antisymmetrically across hemispheres. The remaining two columns of $\hat{W}^{\text{in}}$ — carrying the initial-heading impulse $(\cos \theta_0, \sin \theta_0) \delta_{t,0}$ — remain free for all neurons. The learnable parameters are

$$\{\hat{S}, v_{\text{PENa}_L}, v_{\text{PENa}_R}, v_{\text{PENb}_L}, v_{\text{PENb}_R}, \hat{W}_{:, \theta_0}^{\text{in}}, \hat{b}, \hat{W}^{\text{out}}, \hat{b}^{\text{out}}\};$$

the connectome template W^{con} , PEN subpopulation membership, neuron-type assignments, τ and Δt are given. Training optimises a per-frame MSE on the heading readout with two connectome-prior regularisers, an ℓ_1 sparsity prior on the recurrent magnitudes, and a circular total-variation prior on the EPG ring firing rates,

$$\mathcal{L} = \frac{1}{BT} \sum_{b,t} \|\hat{y}_{b,t} - y_{b,t}\|_2^2 + \lambda_{\text{cos}} \mathcal{L}_{\text{cos}} + \lambda_{\text{nf}} \mathcal{L}_{\text{nf}} + \lambda_1 \|\hat{S}\|_1 + \lambda_{\text{tv}} \mathcal{L}_{\text{tv}}, \quad (5)$$

with block losses computed over type-pair masks $\mathcal{B}_p \subset \{1, \dots, N\}^2$ (one block per ordered pre/post neuron-type pair $p \in \mathcal{P}$),

$$\mathcal{L}_{\text{cos}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \left[1 - \frac{\langle \hat{W}_{\mathcal{B}_p}^{\text{rec}}, W_{\mathcal{B}_p}^{\text{con}} \rangle}{\|\hat{W}_{\mathcal{B}_p}^{\text{rec}}\| \|W_{\mathcal{B}_p}^{\text{con}}\|} \right], \quad (6)$$

$$\mathcal{L}_{\text{nf}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \text{ReLU}(\kappa - \|\hat{W}_{\mathcal{B}_p}^{\text{rec}}\|)^2, \quad (7)$$

$$\mathcal{L}_{\text{tv}} = \frac{1}{BT} \sum_{b,t} \sum_{i=1}^{|\mathcal{R}|} \left| \sigma(\hat{h}_{\pi_{\mathcal{R}}(i+1)}(t)) - \sigma(\hat{h}_{\pi_{\mathcal{R}}(i)}(t)) \right|, \quad (8)$$

where $\pi_{\mathcal{R}}$ is the EPG ring permutation that orders the EPG neurons by their PB glomerulus / preferred-phase index around the ellipsoid body, indices are interpreted modulo $|\mathcal{R}|$ to close the ring, and σ is the sigmoid firing-rate non-linearity of Eq. (1). The cosine-distance term $\mathcal{L}_{\text{cos}}$ anchors each block to the connectome template; the norm-floor term $\mathcal{L}_{\text{nf}}$ prevents any type-pair block from collapsing to zero by penalising the (squared) shortfall of the block mean absolute weight $\|\hat{W}_{\mathcal{B}_p}^{\text{rec}}\|$ below a fixed floor κ . The threshold κ is a hyperparameter (typical value $\kappa = 0.05$) that sets the minimum functional magnitude each type-pair block must retain; because $\mathcal{L}_{\text{cos}}$ is scale-invariant, $\mathcal{L}_{\text{nf}}$ is required to prevent a degenerate solution in which the cosine prior is satisfied by an arbitrarily small recurrent magnitude. The circular total-variation term $\mathcal{L}_{\text{tv}}$ is a *prior on the solution* rather than a constraint imposed by the path-integration task: the ring permutation $\pi_{\mathcal{R}}$ orders the 46 EPG nodes by their PB-glomerulus / EB-column index, so $\mathcal{L}_{\text{tv}}$ penalises spatial jumps in the activity *across EB columns* that the EPG arbors tile, not jumps across the cortical EPG somata (which sit outside the neuropil envelope and are not where the bump is measured). Calcium imaging records GCaMP fluorescence from EPG arbors in the EB and shows that the EPG population forms a *single* coherent bump that moves smoothly around the EB as the fly turns (Seelig and Jayaraman, 2015; Kim et al., 2017; Green et al., 2017); $\mathcal{L}_{\text{tv}}$ shapes the same EB-column profile that this imaging measures, propagating that anatomical-consistency prior to the rest of the network via the recurrent dynamics. Empirically the prior also tightens the trained solutions away from the connectome’s null directions: under the per- $(i, \alpha, \text{instance})$ sparsification of Sect. 3, the predecessor no-TV recipe admits an OLS-calibrated single-edge collapse at rollout $r \approx 0.98$, whereas the present TV-trained CV folds resist that collapse ($r = 0.25 \pm 0.28$); a paired ablation with $\lambda_{\text{tv}} = 0$ on the same recipe (configs `drosophila_cx_pi_epg_no_tv` and `drosophila_cx_pi_gnn_epg_no_tv`) re-runs this contrast under matched seeds. A linear-reconstruction analysis of the null space of the trained $\hat{W}^{\text{rec}}$ — which feeds the per-cell-type and per-unit sparsification tests in the Results — is given in Sect. 3.

Graph neural network model. The GNN model replaces the dense recurrent operator $\sum_j \hat{W}_{ij}^{\text{rec}} \sigma(\hat{h}_j)$ of Eq. (1) with a message-passing graph neural network (Gilmer et al., 2017), and — crucially — does *not* hard-code the leak or the firing-rate nonlinearity. Whereas the RNN imposes the full ODE of Eq. (1), the GNN imposes only the connectome topology and the input/output channels; the leak $-\hat{h}_i$ and the rate nonlinearity σ must be *discovered* inside the learned drift. Nodes correspond to neurons and edges to the non-zero entries of the hemibrain connectome template W^{con} . The GNN learns a latent embedding $\mathbf{a}_i \in \mathbb{R}^2$ per neuron, representing compact cell-type-specific identity, and the subthreshold activity evolves as

$$\tau \frac{d\hat{h}_i(t)}{dt} = f_\theta(\hat{h}_i(t), \mathbf{a}_i, \sum_{j \in \mathcal{N}_i} \hat{W}_{ij} g_\phi(\hat{h}_j(t), \mathbf{a}_j)) + \sum_l \hat{W}_{il}^{\text{in}} u_l(t), \quad (9)$$

with the same readout (Eq. (2)), PEN-gated encoder (Eq. (4)), and dynamics constants $(\tau, \Delta t)$ as the RNN. The neighbour set is fixed by the connectome topology, $\mathcal{N}_i = \{j : W_{ij}^{\text{con}} \neq 0\}$. The node update f_θ and edge message g_ϕ are three-layer perceptrons (width 64, ReLU, linear output). g_ϕ maps the presynaptic state $\hat{h}_j$ to a per-edge message conditioned on the latent embedding $\mathbf{a}_j$ and weighted by the per-edge gain $\hat{W}_{ij}$; f_θ then processes the postsynaptic state $\hat{h}_i$, the embedding $\mathbf{a}_i$, and the aggregated synaptic drive. We constrain g_ϕ to be monotonically increasing in $\hat{h}_j$ via the prior $\partial g_\phi / \partial \hat{h} \geq 0$ (see Eq. (11)); combined with the sign-locked $\hat{W}_{ij}$ (Eq. (10)) this restores Dale’s law at the per-edge level without introducing a $\pm$ -sign degeneracy. The leak, the firing-rate nonlinearity, and any departure from a linear synapse must emerge from $(f_\theta, g_\phi, \mathbf{a})$ themselves; the monotonicity priors below provide the minimum inductive bias for f_θ, g_ϕ to behave like a neuron (postsynaptic drive increases with net excitatory input). The per-edge weight is sign-locked to the connectome as in the RNN,

$$\hat{W}_{ij} = |W_{ij}| \text{sign}(W_{ij}^{\text{con}}) (1 - \delta_{ij}), \quad (10)$$

where $W_{ij} \in \mathbb{R}$ is the trainable per-edge magnitude. The learnable parameters are

$$\{\theta, \phi, \mathbf{a}, W, v_{\text{PENa}_L}, v_{\text{PENa}_R}, v_{\text{PENb}_L}, v_{\text{PENb}_R}, \hat{W}_{:, \theta_0}^{\text{in}}, \hat{b}, \hat{W}^{\text{out}}, \hat{b}^{\text{out}}\};$$

the connectome template W^{con} , PEN subpopulation membership, neuron-type assignments, $\tau, \Delta t$, and ring orderings are given.

As with the RNN, heading recovery from cos/sin readouts is fundamentally non-unique: many distinct $(\theta, \phi, \mathbf{a}, W)$ configurations produce identical readouts. We constrain the solution space with the same connectome-prior regularisers as the RNN ($\mathcal{L}_{\text{cos}}, \mathcal{L}_{\text{nf}}$ from Eqs. (6) and (7), computed on the per-edge gain surface $\hat{W}_{ij}$ with floor threshold κ), the same circular-TV calcium-imaging prior $\mathcal{L}_{\text{tv}}$ on the EPG ring firing rates (Eq. (8)), and sparsity and monotonicity priors on the MLPs:

$$\begin{aligned} \mathcal{L} = & \frac{1}{BT} \sum_{b,t} \|\hat{y}_{b,t} - y_{b,t}\|_2^2 + \lambda_{\text{cos}} \mathcal{L}_{\text{cos}} + \lambda_{\text{nf}} \mathcal{L}_{\text{nf}} + \lambda_{\text{tv}} \mathcal{L}_{\text{tv}} \\ & + \lambda_0 \|\theta\|_1 + \lambda_1 \|\phi\|_1 + \lambda_2 \|\theta\| + \lambda_3 \|\phi\| + \gamma_0 \|W\|_1 + \gamma_1 \|W\| \\ & + \mu_0 \left\| \text{ReLU} \left(+ \frac{\partial f_\theta(\hat{h}, \mathbf{a}, m)}{\partial \hat{h}} \right) \right\| + \mu_1 \left\| \text{ReLU} \left(- \frac{\partial f_\theta(\hat{h}, \mathbf{a}, m)}{\partial m} \right) \right\| \\ & + \mu_2 \left\| \text{ReLU} \left(- \frac{\partial g_\phi(\hat{h}, \mathbf{a})}{\partial \hat{h}} \right) \right\| + \mu_3 \|g_\phi(h_\star, \mathbf{a}) - h_\star\|, \end{aligned} \quad (11)$$

where $m = \sum_{j \in \mathcal{N}_i} \hat{W}_{ij} g_\phi(\hat{h}_j, \mathbf{a}_j)$ is the aggregated synaptic input. The ℓ_1 penalties λ_0, λ_1 on the MLP weights bias f_θ, g_ϕ toward sparser, simpler functions; the ℓ_2 penalties λ_2, λ_3 prevent weight explosion. The leak-monotonicity term μ_0 penalises *positive* slopes of f_θ in the postsynaptic state $\hat{h}$,

encoding the prior that the drift contains a leak ($\partial f_\theta / \partial \hat{h} \leq 0$); μ_1 penalises negative slopes of f_θ in the aggregated input m , encoding the prior that postsynaptic drive increases with net excitatory input. Together, μ_0 and μ_1 force the discovered drift to be decreasing in self-voltage and increasing in synaptic input — the two qualitative signatures of Eq. (1). The term μ_2 enforces the analogous monotonicity prior on g_ϕ in the presynaptic voltage. The normalisation term μ_3 anchors g_ϕ at a reference subthreshold value h_* , removing the multiplicative degeneracy between g_ϕ and $\hat{W}$ in Eq. (9). Hyperparameters $\lambda_{\{0,1,2,3\}}$, $\gamma_{\{0,1\}}$, $\mu_{\{0,1,2,3\}}$ and the MLP widths/depths are tuned with the agentic optimisation procedure of (Allier et al., 2026a).

Mutual-information estimators. We use mutual information to quantify how strongly each unit’s firing rate $r_i(t) = \sigma(\hat{h}_i(t))$ tracks the heading $\theta(t)$ (or the angular velocity $\omega(t)$) over a held-out rollout. For a single neuron we wrap θ to $(-\pi, \pi]$ via $\theta \leftarrow \arg(e^{i\theta})$, discretise the wrapped angle into n_θ equal-width angular bins and r into n_r equal-width bins, then compute the plug-in estimator in bits,

$$\hat{I}(r_i; \theta) = \sum_{b,c} \hat{p}(b, c) \log_2 \frac{\hat{p}(b, c)}{\hat{p}(b) \hat{p}(c)}, \quad (12)$$

where $\hat{p}$ is the empirical joint over the $(\theta$ -bin, r -bin) histogram. For a group of K neurons we use the cross-validated lower bound

$$\hat{I}_{\text{LB}}(\mathbf{r}_{1:K}; \theta) = H(\theta) - \frac{1}{|\mathcal{D}_{\text{te}}|} \sum_{(t, \theta) \in \mathcal{D}_{\text{te}}} -\log_2 q_\psi(\theta_b \mid \mathbf{r}(t)), \quad (13)$$

where q_ψ is an L_2 -regularised multinomial logistic regression with input $\mathbf{r} \in \mathbb{R}^K$ and output the n_θ heading bins, trained on a stratified 5-fold split. $\hat{I}_{\text{LB}}$ is the population-MI lower bound used for the per-cell-type and all-neurons joint MIs in Figs. 13 and 14 and Tabs. 1 and 2. The same estimators apply to ω by replacing θ with ω and using uniform binning over the 99th percentile of $|\omega|$ instead of circular wrapping.

Agentic hyperparameter optimisation. The GNN hyperparameter space (embedding dimension, MLP widths and depths, the λ , γ , μ coefficients, learning rates, and the curriculum on the rollout length T) is too large for grid search; we tune it with the closed-loop LLM-agent protocol of (Allier et al., 2026a), using the deterministic-sweep rollout correlation $r_{\text{roll},1k}$ (Pearson r on the decoded heading at $\omega = 60^\circ/\text{s}$, $T = 1000$ frames, warm-up 10) as the primary score. The first architectural promotion off the stabilisation winner was $\dim \mathbf{a}_i : 2 \rightarrow 4$, with best-seed $r_{\text{roll},1k} = 0.981$ and 5-seed mean 0.931 ± 0.061 versus a seed-fragile $\dim \mathbf{a}_i = 2$ parent. The same four-dimensional latent recurs in the connectome-recovery agentic explorations of (Allier et al., 2026a): the noise-free condition deploys $\dim \mathbf{a} = 4$, the $\sigma = 0.05$ conditions saturate at 2–4, and $\dim \mathbf{a} \geq 6$ falsifies on connectivity- R^2 . A four-dimensional embedding appears to be the smallest representation that separates the dozen-plus presynaptic cell-type identities g_ϕ must encode without over-fitting per-edge gains. The runs of Fig. 7 predate this promotion and were trained at $\dim \mathbf{a}_i = 2$.

Why the agentic loop is methodologically primordial. In high-dimensional recurrent systems, optimisation failure is not itself strong evidence against a mechanistic hypothesis. The non-convergence of a parameterisation may reflect poor hyperparameter selection, unstable training dynamics, or an unfavourable basin of the optimisation landscape rather than the absence of a valid solution. The agentic optimisation loop is therefore not merely a performance tool but a methodological requirement: without extensive closed-loop exploration of the training regime, it is easy to mistake optimisation failure for biological falsification.

Two-stage agentic protocol. We run each parameterisation through *two* agentic loops with distinct objectives. The first loop is goal-driven: it searches the hyperparameter space until $r_{\text{roll},1k}$ saturates at the behavioural ceiling, establishing that the parameterisation admits a converging recipe at all. The second loop is knowledge-driven: starting from the first loop’s winner as a known floor, it conducts systematic single-axis sweeps across all hyperparameter families — learning rates, stabilisers, curriculum, regularisation, initialisation, architecture — under a strict causality rule (exactly one parameter changes per slot), interleaved with robustness batches (identical configuration, varied seeds) to separate genuine effects from seed variance. The two-stage split is essential because the first loop produces only the existence proof ("there is a recipe that converges") whereas the second produces the causal map the paper’s interpretive claims rely on: which hyperparameters explain variance in the converged metric, which failure modes the parameterisation exhibits when training breaks, how robust the converged recipe is across seeds. Collapsing the two stages into a single optimisation pass yields a winning configuration without the mechanistic understanding such conclusions require.

3 Results

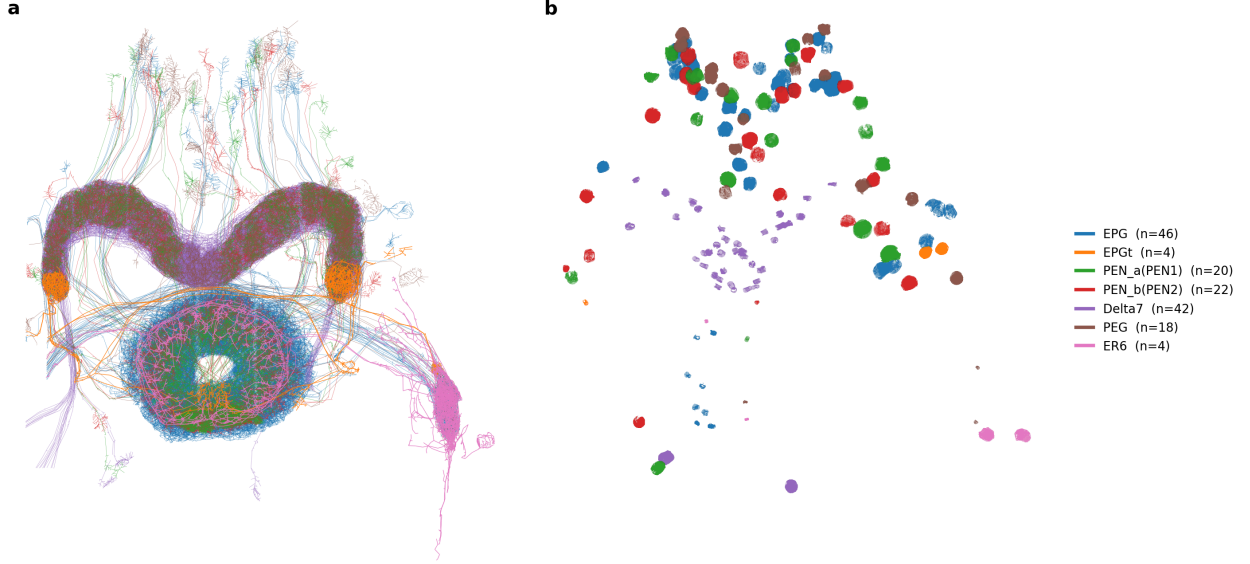


Figure 1: **Anatomical morphology of the 156 hemibrain CX neurons modelled in this paper.** (a) 3-D skeleton render of the seven cell types (EPG, EPGt, PEN_a/PEN_b, $\Delta 7$, PEG, ER6; n per type in the legend) overlaid on the neuropil outlines of EB, PB, FB, NO and BU_{L/R} (grey silhouettes). View flipped so dorsal is up: PB at the top, FB in the middle, EB ring at the bottom, BU_{L/R} on the right. (b) Same view restricted to the cell bodies. Each neuron contributes its soma 3-D mesh, recovered from the hemibrain segmentation by cropping the full per-neuron mesh (**hemibrain-dvid**) to a sphere of radius $1.4 r_{\text{soma}}$ around the soma node identified in the SWC skeleton (the soma is the only node with radius $\gg$ the segmental baseline). The two panels show *spatially disjoint* compartments of the same set of neurons: in (a) the dense ring / moustache / fan structures (EB, PB, FB) are *neuropils* — regions of axo-dendritic arborisation with no cell bodies inside — and in (b) the somata sit in the cortex around and above those neuropils, each connected to its neuropil arbor by a thin primary-neurite tract visible in (a). The EB ring in particular is therefore not a ring of cell bodies but the toroidal neuropil substrate the EPG / PEN / $\Delta 7$ arbors tile to support the ring-attractor dynamics. In (a), the EPG axons terminate in the PB (top, 18 glomeruli per the L1–L9 / R1–R9 numbering of Hulse et al. (2021)) and their dendrites tile the 16 wedges of the EB ring (centre); $\Delta 7$ arborises across multiple PB glomeruli; ER6 lies largely outside the CX core (right). Cell bodies (b) sit in the cortex with no overlap with the neuropils in (a). Cortical cell bodies of each population sit outside the neuropil envelope in a stereotyped pattern — e.g. the four ER6 somata cluster in one anatomically distinct location while the 42 $\Delta 7$ somata wrap above the PB. The same colour assignment is used for the per-cell-type bars in Figs. 13 and 14 and the row blocks in Fig. 2, so the morphology, connectivity and HD coding panels can be read against each other neuron type by neuron type.

We compare four recurrent parameterisations on the same path-integration task, each trained on the OU velocity stream of Eq. (1) and evaluated on a constant- $\omega = +60^\circ/\text{s}$ deterministic rollout, a held-out OU trial, and an integration-gain sweep over $|\omega| \leq 180^\circ/\text{s}$. (i) **Known-ODE RNN** (Fig. 3). Connectome topology and Dale signs are imposed via $\hat{W}^{\text{rec}} = |\hat{S}| \odot \text{sign}(W^{\text{con}})$; the per-edge magnitudes $\hat{S}$, the four PEN velocity scalars, the encoder/decoder and biases train end-to-end.

Attains $r = 1.00$, $\text{RMSE} = 2.6^\circ$ on the constant- ω rollout, integration gain ≈ 1 across $|\omega| \leq 120^\circ/\text{s}$, and accurately tracks the OU trial. **(ii) Frozen- W_{rec} control** (Fig. 5). Same architecture with $\hat{W}^{\text{rec}} \equiv W^{\text{con}}$ held at the raw hemibrain values; only the encoder, decoder, biases, and PEN scalars train. Fails on every readout ($r = -0.37$, $\text{RMSE} = 105^\circ$, integration gain ≈ 0): the encoder/decoder cannot memorise the path-integration map when the recurrent operator is held at the connectome’s nominal values, confirming that the raw weights do not by themselves instantiate a continuous attractor. **(iii) Fully connected RNN** (Fig. 6). The connectome topology mask is dropped; only the column-wise Dale sign of each presynaptic neuron is enforced. Recovers heading accuracy ($r = 1.00$, $\text{RMSE} = 12.4^\circ$) but populates off-block entries that are absent from the hemibrain wiring (Fig. 6b). **(iv) Message-passing GNN** (Fig. 7). The dense recurrent operator is replaced by f_θ, g_ϕ MLPs over a per-neuron embedding $\mathbf{a}_i$ (Eq. (9)); the leak and firing-rate non-linearity must emerge from these MLPs. The checkpoint shown is the epoch-5 best model of the soft-curriculum tail-loss training, reaching $r_{\text{roll},1k} \approx 0.99$ on the deterministic constant- ω rollout and tracking the held-out OU trial to within the same band as the RNN variants ($\text{RMSE} = 17.0^\circ$ on the constant- ω rollout).

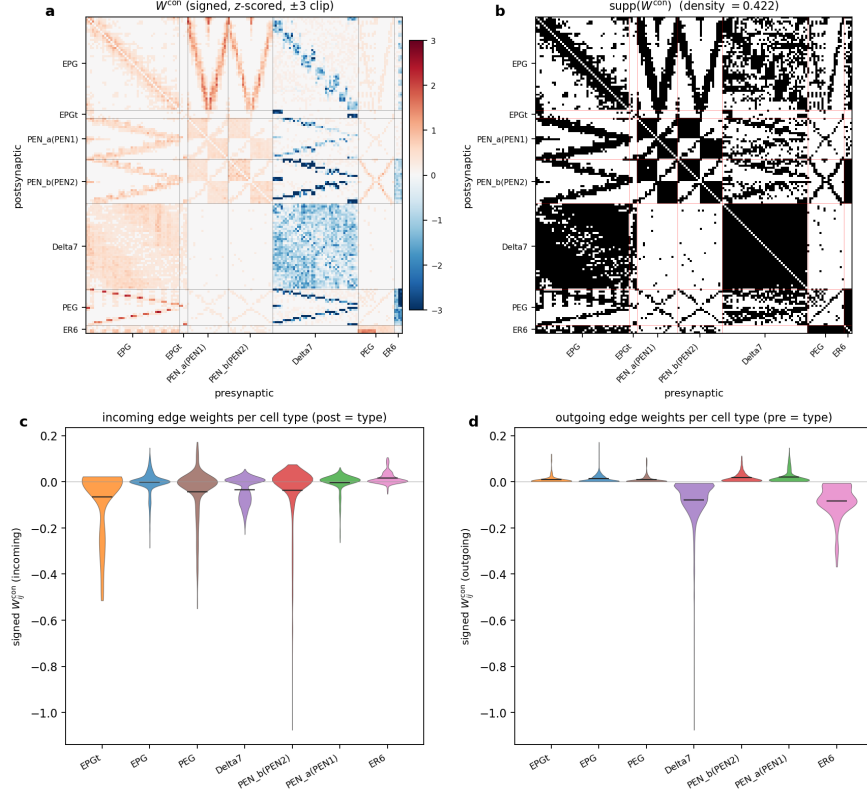


Figure 2: CX connectome W^{con} : structure and per-cell-type edge-weight distributions. The matrix is the effective connectome the trained Known-ODE / GNN models see, derived from the hemibrain export (*papers/Code_NN/Code_NN/Data/Figure5/exported-traced-adjacencies-v1.2/*, files *traced-neurons.csv* and *traced-total-connections.csv*); it follows the Hulse-2025 preparation in which $\Delta 7$ and ER6 outputs are sign-flipped (inhibitory) and ER6 outputs are amplified $5\times$ before W^{con} is normalised to spectral radius 0.9 (Eq. (3), Sect. 2). **(a)** Signed W^{con} , z-scored over non-zero entries and clipped to ± 3 ; neuron-type blocks marked on both axes; red = excitatory, blue = inhibitory. **(b)** Binary support $\text{supp}(W^{\text{con}})$ (black = edge present, white = absent); the matrix is sparse (density 0.42, 10 263 non-zero edges out of $156^2 = 24\,336$). **(c)** Per-cell-type *incoming* edge-weight distributions (violin, mean marked): for each type t , the data are the non-zero W_{ij}^{con} with the post-synaptic neuron i in t (variation across all pre-synaptic partners j). **(d)** Per-cell-type *outgoing* edge-weight distributions: for each type t , the data are the non-zero W_{ij}^{con} with the pre-synaptic neuron j in t (variation across all post-synaptic partners i). Dale’s law is visible in (d) as one-sided violins: every type is uniformly excitatory except $\Delta 7$ and ER6, whose outgoing distributions sit entirely below zero. Cell-type ordering and colour assignment match Figs. 13 and 14, so the rows there can be read off directly against the connectome statistics here.

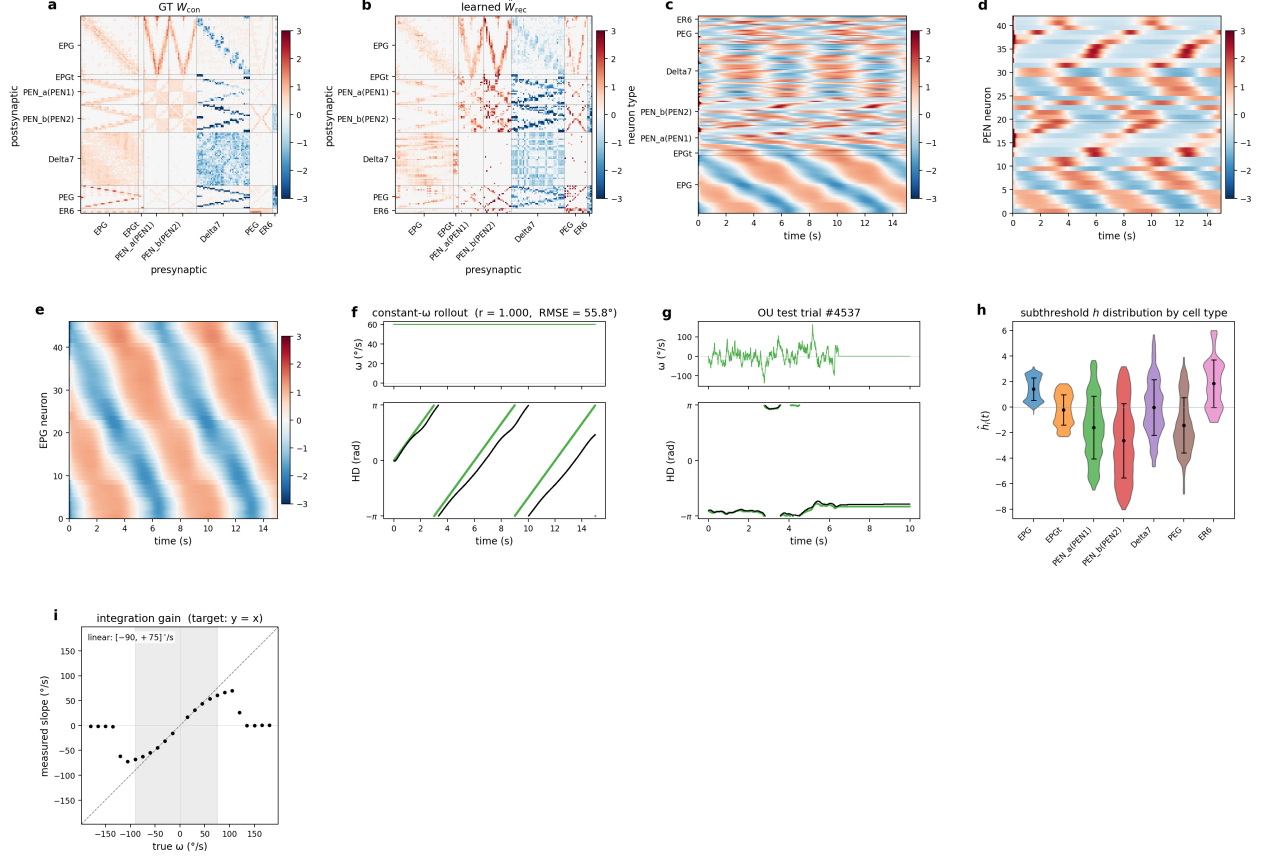


Figure 3: Known-ODE RNN: connectome topology and Dale signs imposed, per-edge magnitudes learned. Sign-locked RNN with $\hat{W}^{\text{rec}} = |\hat{S}| \odot \text{sign}(W^{\text{con}})$, trained for 10 epochs. **(a)** GT connectome W^{con} , z-scored over non-zero entries and clipped to ± 3 ; neuron-type blocks marked on both axes. **(b)** Learned $\hat{W}^{\text{rec}}$ on the same scale. **(c)** All 156 neurons over a constant- $\omega = +60^\circ/\text{s}$ rollout, sorted by cell type; firing rates z-scored per neuron. **(d)** PEN sub-population over the same rollout. **(e)** EPG sub-population over the same rollout. **(f)** Constant- ω rollout. Top: $\omega(t)$. Bottom: true heading (green) and decoded heading (black), wrapped. Title reports post-warmup Pearson r and circular RMSE. **(g)** Same layout on a held-out OU test trial. **(h)** Per-cell-type distribution of the subthreshold state $\hat{h}_i(t)$ (violin + mean ± 1 SD), revealing where each population's activity sits relative to the sigmoid's linear regime. **(i)** Integration gain: linear-fit slope of the unwrapped decoded heading vs true ω , on a $15^\circ/\text{s}$ grid over $[-180, +180] \setminus \{0\}$. Dashed line: $g = 1$; grey band marks the linear-tracking domain.

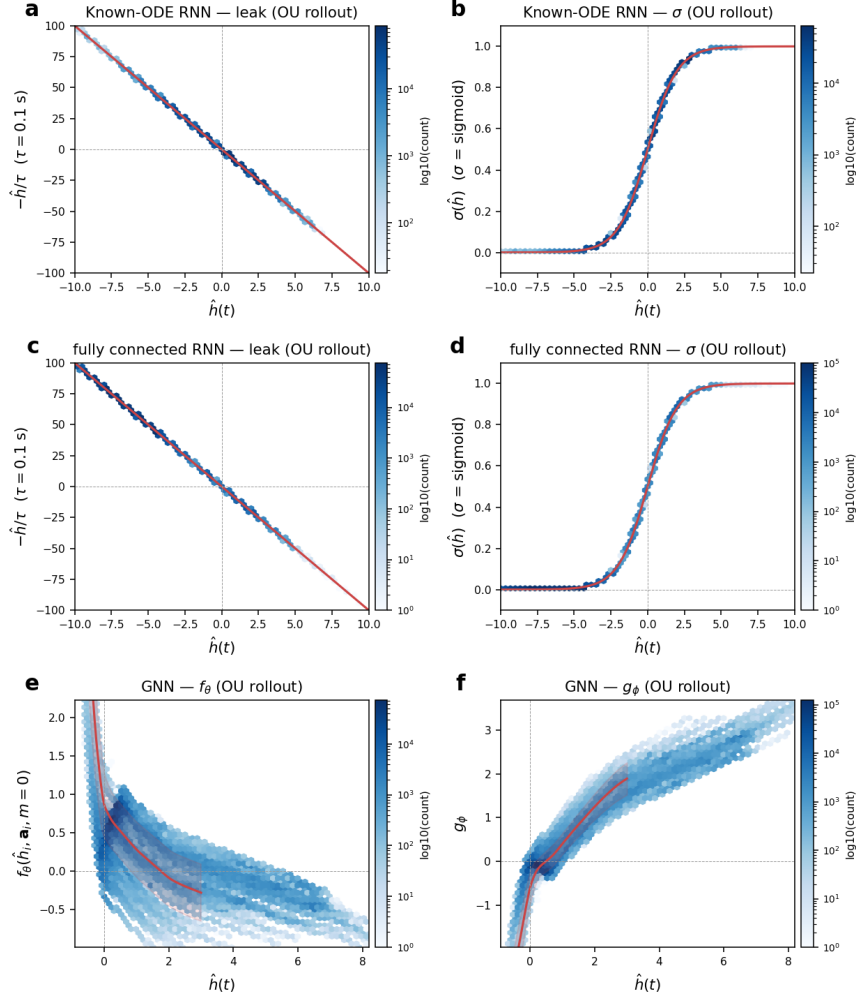


Figure 4: **Operating range of the leak/drift and the firing-rate non-linearity along a natural OU velocity rollout** ($T = 10\,000$ frames) for the three converged CX models (the frozen- W^{rec} control fails to form a bump and is omitted here). Hexbin density (log10 count, Blues) of the per-(neuron, time) samples; red curve is the static function over the plotted $\hat{h}$ range (with ± 1 SD across the 156 neurons where applicable). **(a, b)** Known-ODE RNN (Sect. 2): hardcoded leak $-\hat{h}/\tau$ ($\tau = 0.1$ s) and sigmoid $\sigma(\hat{h})$. The rollout sweeps the full $|\hat{h}| \lesssim 10$ domain — the bump traverses every glomerular phase as heading changes, painting the entire linear leak and saturating σ in both tails. **(c, d)** fully connected RNN (only column-wise Dale signs enforced, topology mask dropped): same hardcoded $-\hat{h}/\tau$ and σ as (a, b), but the operating range visited is shaped by a different learned $\hat{W}^{\text{rec}}$ that populates off-block entries absent from the hemibrain wiring. **(e, f)** GNN with the Dale-conformant $\partial g_\phi / \partial v \geq 0$ monotonicity prior: the operating range concentrates in a narrow $\hat{h} \in [-2, 1]$ band near a flat plateau, with the steep non-linear regimes of f_θ and g_ϕ sparsely probed by the bump-encoding sub-population.

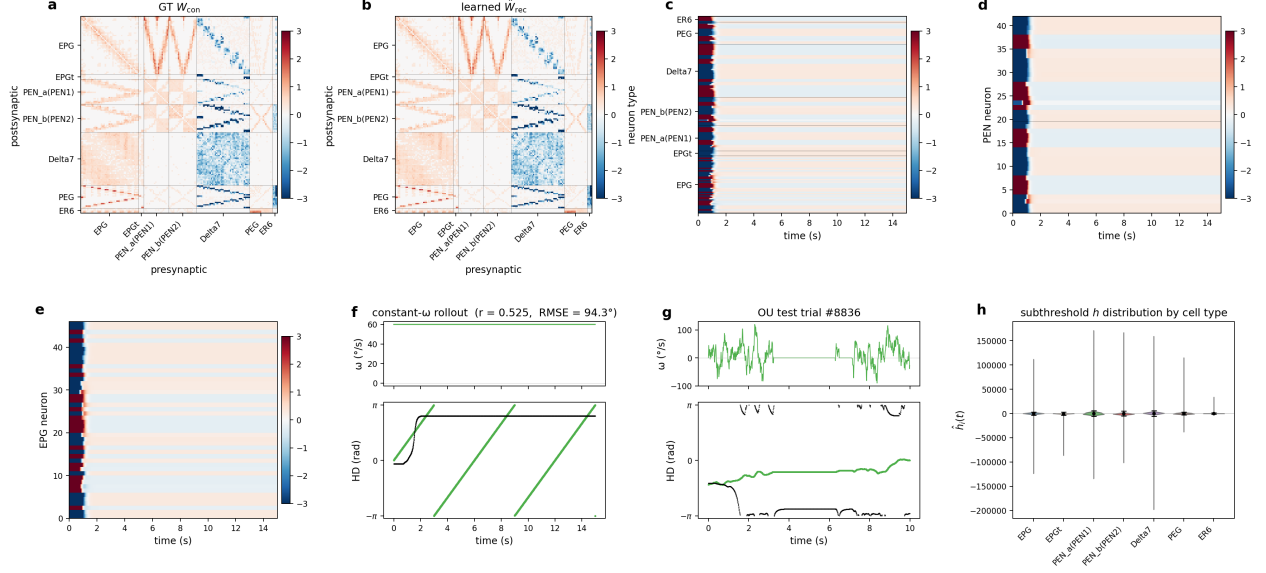


Figure 5: **Frozen- W_{rec} control.** Panels follow Fig. 3. The recurrent matrix is held at $\hat{W}^{rec} \equiv W^{con}$ throughout training; only $\hat{W}^{in}$, $\hat{W}^{out}$, biases, and the four PEN velocity scalars train. The decoded heading is flat (panel (f); $r = -0.37$, $RMSE = 105^\circ$), the network does not respond to the OU velocity input (panel (g)), and the integration gain is zero across the full ω range (panel (i)). The model fails to form a bump, so the tuning, phase-shift, and FWHM panels (j-l) are omitted.

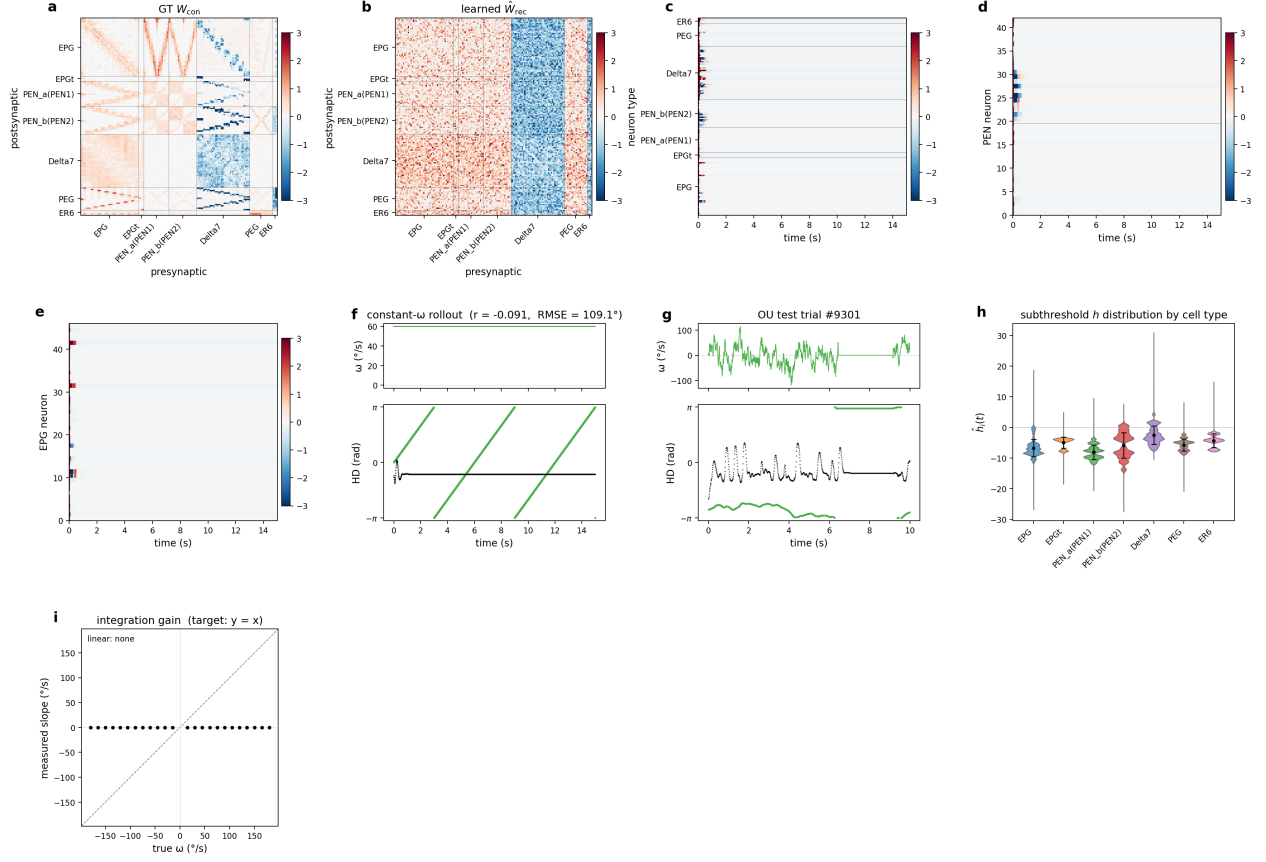


Figure 6: **Fully connected RNN**. Panels follow Fig. 3. The connectome topology mask is dropped; $\hat{W}_{rec} = |\hat{S}| \odot \text{sign}(\sum_i W_{i,:}^{con})$ enforces only the dominant E/I sign per presynaptic neuron and the recurrent matrix is otherwise dense. The learned matrix in (b) carries weight outside the hemibrain edge support; heading accuracy is preserved ($r = 1.00$, RMSE = 12.4°).

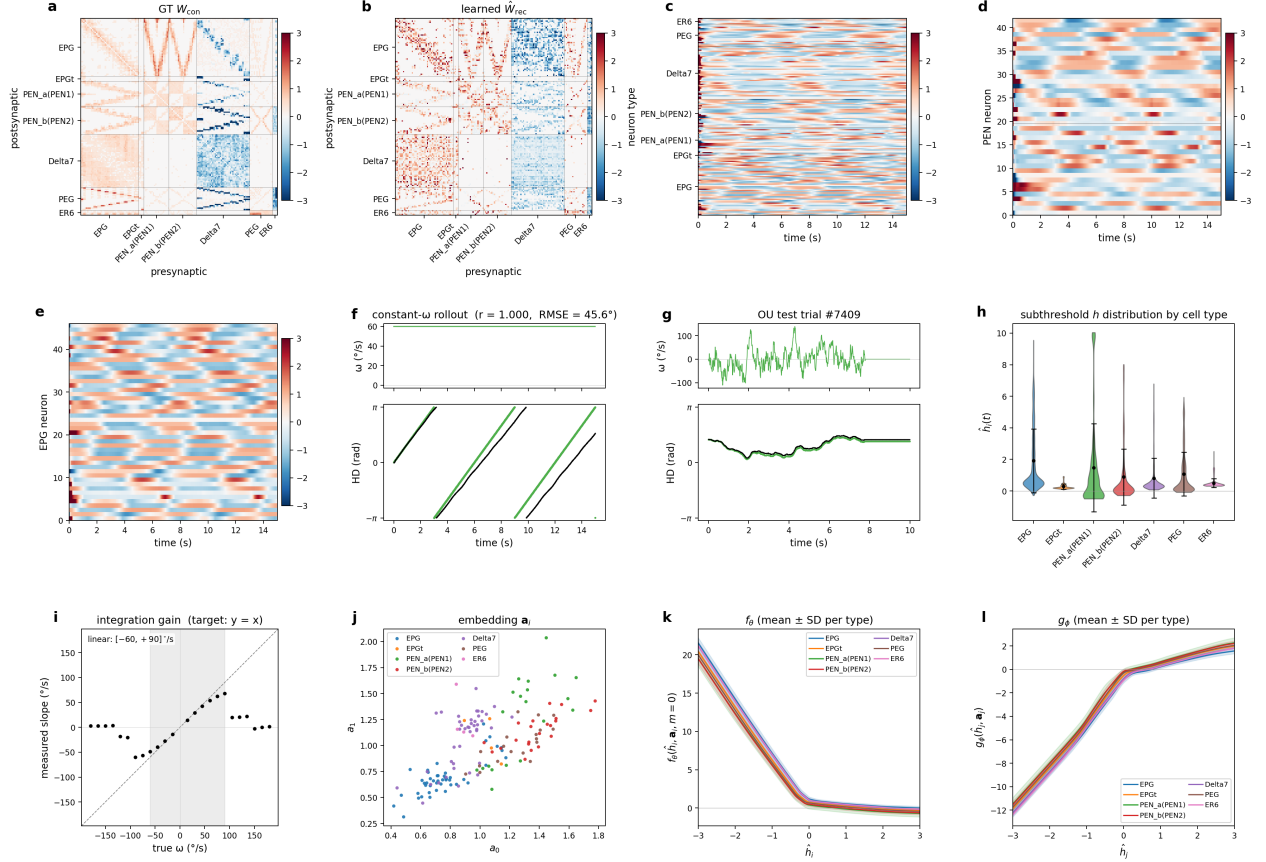


Figure 7: **Message-passing GNN (Sect. 2).** Panels (a)–(i) follow Fig. 3. The dense recurrent operator is replaced by $f_\theta(\hat{h}_i, \mathbf{a}_i, \sum_j \hat{W}_{ij} g_\phi(\hat{h}_j, \mathbf{a}_j))$ (Eq. (9)); the leak and firing-rate non-linearity emerge from f_θ, g_ϕ . Panel (b) shows the per-edge gain $\hat{W}_{ij}$ at the connectome edge positions (the GNN does not form a dense $\hat{W}^{\text{rec}}$). (j) Per-neuron latent embedding $\mathbf{a}_i \in \mathbb{R}^2$, coloured by cell type. (k) Learned $f_\theta(\hat{h}_i, \mathbf{a}_i, m=0)$ for each neuron, plotted over its activity domain $[\mu_i - 2\sigma_i, \mu_i + 2\sigma_i]$ (snapshot from training). Per the GNN ODE (Eq. (9)), f_θ should approximate $-\hat{h}_i/\tau$ to recover the leak. (l) Learned $g_\phi(\hat{h}_j, \mathbf{a}_j)$ over the same per-neuron domain (snapshot from training) — the per-edge message that enters the GNN aggregation. The per-edge signed drive is $\hat{W}_{ij} g_\phi(\hat{h}_j, \mathbf{a}_j)$ (single-power, no squaring), so the effective synaptic sign is the product $\text{sign}(\hat{W}_{ij}) \cdot \text{sign}(g_\phi)$: Dale’s law is no longer enforced per edge, though the dominant E/I row pattern is still encoded in $\hat{W}$. Because the squaring/positivity constraint is dropped, the $\pm$ sign degeneracy on g_ϕ is removed and the recovered firing-rate non-linearity is uniquely identified — here sigmoidal in the standard “depolarised-is-active” convention, recovering the firing-rate non-linearity of Eq. (1). Checkpoint shown is the epoch-5 best model of the EPG-readout tail-loss training.

Known-ODE RNN: many circuits, same activity. The path-integration task fixes only the input–output map: angular velocity $\omega(t)$ in, decoded heading $\hat{\theta}(t)$ out. The Known-ODE RNN (Sect. 2, Fig. 3) imposes the full ODE of Eq. (1) — linear leak $-\hat{h}/\tau$, sigmoid firing-rate non-linearity, hemibrain connectome topology + Dale signs on $\hat{W}^{\text{rec}}$ — and trains only the per-edge magnitudes $\hat{S}$. Across 10 independently initialised CV folds (per-fold randomised simulation, training, and task seeds; the winning recipe of Sect. 2), every fold converges to $r_{\text{roll},1k} = 1.000 \pm 0.000$ at $T = 1000$ frames. The recovered $\hat{h}$ trajectories cover both saturating tails of the sigmoid (Fig. 4a, b: hexbin

density of $(\hat{h}, \sigma(\hat{h}))$ over a $T = 10,000$ frame OU rollout shows non-zero occupancy out to $|\hat{h}| \approx 10$, with the bulk of the per-neuron mass spread across the linear regime).

The 10 converged $\hat{W}^{\text{rec}}$ matrices reach the same behavioural ceiling but *not* the same recurrent weights (Fig. 8). The pairwise cosine similarity over the connectome support (panel a) is 0.71 ± 0.02 between learned folds (45 off-diagonal pairs) and 0.63 ± 0.02 against W^{con} : the learned solutions cluster more tightly with each other than with the connectome template they sign-inherit from, but the inter-fold spread is still substantial. Panel b confirms this at the per-cell-type-block level: some connection blocks are tightly constrained across folds (median coefficient of variation $\lesssim 0.5$, e.g. $\Delta 7 \rightarrow \text{PEN}_b$, $\text{PEN}_b \rightarrow \text{ER6}$), while others are essentially free to wander (CV $\gtrsim 1$, e.g. $\Delta 7 \rightarrow \Delta 7$, $\text{EPGt} \rightarrow \text{PEN}_{a/b}$). Panel c shows that along the connectome support the trained matrices are systematically 2–50 $\times$ larger than W^{con} (mean $\log_2$ ratio positive for every block), and that this amplification pattern is itself reproducible across folds ($\log_2$ -ratio SDs $\lesssim 0.5$): training rescales the hemibrain template along its existing edges rather than carving a different support.

Behaviour therefore does *not* identify the per-edge weights: the 10 folds reach the same $r_{\text{roll},1k}$ via different recurrent matrices that share an overall magnitude profile but diverge at the per-edge level, and Sect. 3 shows that per-cell-type partitions of the connectome support absorb sum-zero perturbations at moderate amplitude ($\bar{r} = 0.99 \pm 0.05$ at $\lambda = 1$ across 10 models). Many circuits, one activity. Tightening this equivalence class further would require pinning the dynamics to more than just the input–output map: per-neuron activity recordings across the EPG / PEN / $\Delta 7$ populations would constrain the voltage trajectories themselves and collapse the per-edge ambiguity that the heading readout alone cannot resolve.

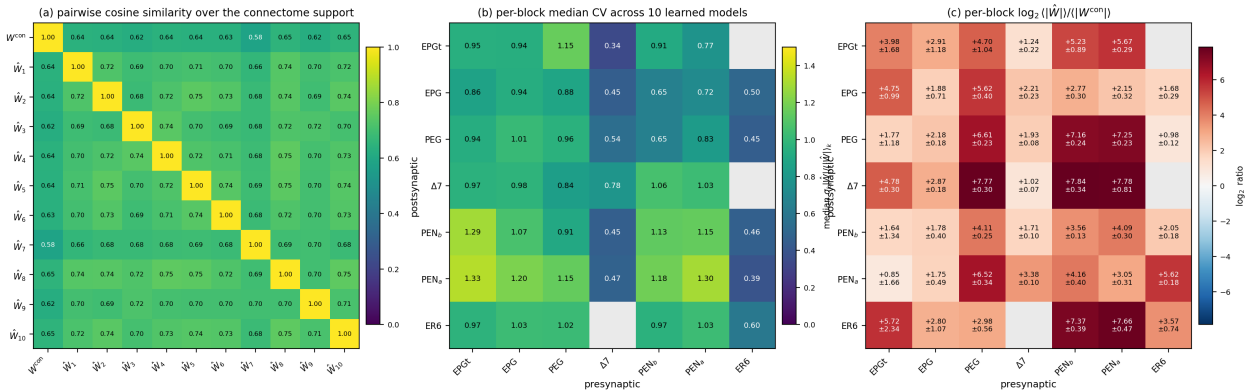


Figure 8: **Comparison of the 10 trained Known-ODE $\hat{W}^{\text{rec}}$ to each other and to the connectome template W^{con} .** (a) 11 $\times$ 11 pairwise cosine similarity over the connectome support, ordered W^{con} first, then 10 CV folds. Diagonal = 1.0. Off-diagonal in the bottom-right 10 $\times$ 10 block is 0.71 ± 0.02 (inter-fold); first row / column off-diagonal is 0.63 ± 0.02 ($\hat{W}_k$ vs. W^{con}). Learned solutions cluster more tightly with each other than with the connectome template. (b) Per-block median coefficient of variation across the 10 folds, $\sigma_k|\hat{W}_k|/\langle|\hat{W}_k|\rangle_k$, for each (post-type, pre-type) cell-type pair on the same 7 $\times$ 7 block layout as Fig. 2c, d. Low CV = tightly constrained block; high CV = free block. Empty grey cells have no connectome edges between that cell-type pair. (c) Per-block $\log_2$ ratio of the mean learned magnitude to the mean connectome magnitude, $\log_2 [\langle|\hat{W}_k|\rangle/\langle|W^{\text{con}}|\rangle]$, annotated as $\mu \pm \text{SD}$ across the 10 folds. All blocks are uniformly positive (the trained solutions amplify the connectome along its existing support), with the amplification factor varying by an order of magnitude across cell-type pairs and tightly conserved across folds.

To probe how HD information is parcellated across cell types, we estimate the mutual information

$I(\sigma(\hat{h}_i); \theta)$ between each neuron’s rate and heading along a $T = 10\,000$ frame OU rollout ($n_\theta = 64$ angular bins; Sect. 2, Fig. 13, Tab. 1). The whole-population joint MI is $I_{\text{pop}} = 3.43$ bits, predicting a Gaussian-channel per-step posterior $\sigma_{\theta|\hat{h}} \approx \sigma_\theta 2^{-I_{\text{pop}}} \approx 9.6^\circ$ — in the same band as the directly measured circular RMSE (2.5° constant- ω , 25.8° OU; the rollout error compounds integration drift on top of the per-step decoding floor), confirming the MI estimator tracks the network’s actual heading precision. The per-neuron champion is EPGt ($\bar{I} = 1.54$ bits) and ER6 carries essentially no HD ($\bar{I} < 0.01$ bits), so in this solution ER6 acts as a tonic inhibitor without θ -dependent modulation. Per-cell-type cloning is summarised by the redundancy factor $\rho_t = \sum_{i \in t} I_i / I_{\text{joint},t}$ (Tab. 1); large ρ means same-type cells carry near-identical θ -tuning, so per-neuron MIs sum sub-additively into the joint. The EPG block has $\rho \approx 18$ (46 cells $\rightarrow$ 16 effective glomeruli, joint saturating against $H(\theta)$), which also explains why the all-neurons joint MI (~ 3.4 bits) is so much smaller than $\sum_{t,i} I_i$ (~ 150 bits naively): the code is heavily cloned and capped near the angular-entropy ceiling rather than scaling with neuron count. The remaining three trained parameterisations (Fig. 5, Fig. 6, Fig. 7) expose three further layers of this degeneracy, addressed in turn below.

Frozen- W^{rec} control: the raw connectome alone does not explain it. A natural reading of the Known-ODE RNN’s success is that the connectome itself — not the trained magnitudes — carries the path-integration computation. The frozen- W^{rec} control (Sect. 2 with $\hat{W}^{\text{rec}} \equiv W^{\text{con}}$ held at the raw hemibrain values throughout training, Fig. 5) falsifies that reading. Only the encoder $\hat{W}^{\text{in}}$, the decoder $\hat{W}^{\text{out}}$, the biases, and the four PEN velocity scalars are trainable; the recurrent operator is locked. The model fails on every readout ($r = -0.37$, RMSE = 105° , integration gain ≈ 0), the decoded heading is flat, and no bump forms. The raw hemibrain weights therefore do not by themselves instantiate a continuous attractor in this task and training regime: the learned per-edge magnitudes of the Known-ODE RNN are functionally load-bearing. This rules out the cleanest "connectome alone explains the behaviour" narrative and shifts the question to what the *learned* magnitudes are doing — which the next two parameterisations probe.

Fully connected RNN: connectome zeros are not required for the computation. Replacing the topology mask with a column-wise Dale sign constraint ($\hat{W}^{\text{rec}} = |\hat{S}| \odot \text{sign}(\sum_i W_{i,:}^{\text{con}})$, Fig. 6) allows the recurrent matrix to populate every off-block entry absent from the hemibrain wiring. Despite this, the model still reaches $r = 1.00$ and exhibits the same ring-attractor bump migration as the topology-locked Known-ODE RNN (Fig. 12). In this task and training regime, the connectome’s pattern of missing edges is therefore not required for the recovered heading computation: the absent edges behave as free parameters of the inverse problem. The per-cell-type HD MI (Fig. 13c, d) reaches $I_{\text{pop}} = 3.05$ bits ($\sigma_{\theta|\hat{h}} \approx 12.5^\circ$, again consistent with the rollout RMSE), but the internal allocation diverges from the Known-ODE solution: the optimiser drives the 4 EPGt cells into deep negative saturation ($\bar{h} \approx -8$, $\sigma(\hat{h}) < 0.05$, $\bar{I} = 0.01$ bits) and $\Delta 7$ takes over as the dominant per-neuron encoder ($\bar{I} = 1.15$ bits).

GNN with learned neuronal dynamics: the implementation is not uniquely identified by the task. The GNN (Sect. 2, Fig. 7) keeps the connectome topology mask but replaces the hardcoded leak and sigmoid of the Known-ODE RNN with learned functions f_θ, g_ϕ . Despite this much weaker dynamical prior, the trained GNN still reaches $r_{\text{roll},1k} \approx 0.99$ and reproduces the same ring-attractor behaviour. The recovered dynamics differ substantially from the reference Known-ODE model. The learned drift f_θ (Fig. 7k) is not the linear leak $-h/\tau$, and the learned signalling function g_ϕ (Fig. 7l) does not reproduce the canonical sigmoid except at a coarse qualitative level enforced by the monotonicity prior. Yet the resulting network still performs path integration with

near-identical behavioural accuracy. Moreover, the rollout probes only a narrow operating region of the learned functions (Fig. 4g, h): most neurons remain near a low-activity plateau while only a small bump-encoding sub-population traverses the strongly nonlinear regime. Large regions of the learned dynamical surfaces are therefore effectively unconstrained by the task. The GNN thus exposes a deeper layer of degeneracy than the per-weight or connectome-zero ambiguities of the previous models. The path-integration task does not uniquely identify the neuronal dynamics themselves: multiple drift and signalling functions can realise the same ring-attractor computation. The same degeneracy reaches the per-cell-type HD code (Fig. 13e, f; $I_{\text{pop}} = 3.35$ bits): the learned g_ϕ permits the network to suppress $\Delta 7$ ($\bar{I} = 0.11$ bits) and route HD through ER6 ($\bar{I} = 1.18$ bits) instead — the exact mirror image of the Known-ODE allocation. This $\Delta 7 \leftrightarrow \text{ER6}$ exchange is not a random redistribution: $\Delta 7$ and ER6 are the *only* two inhibitory cell types in the model (Fig. 2d, with both populations carrying sign-flipped, $5\times$ -amplified outgoing weights per the Hulse-2025 preparation; see Sect. 2), and their postsynaptic targets nest — 96% of ER6’s targets are also targeted by $\Delta 7$, both blanketing the EPG / PEN / PEG ring. The two populations are therefore approximately interchangeable as sources of ring inhibition, and the GNN exploits this structural symmetry to satisfy the task by carrying HD on ER6 instead of $\Delta 7$. What ER6 cannot replace is $\Delta 7$ ’s recurrent self-inhibition (the $\Delta 7 \rightarrow \Delta 7$ block is by far the largest single signed drive in W^{con} ; Fig. 2a), and consistent with this the silenced $\Delta 7$ population collapses into 42 redundant flat traces ($\rho \approx 30$; Tab. 1). We can directly test this swap only because $\Delta 7$ and ER6 are the two inhibitory types in the CX preparation; whether analogous exchanges hold among the excitatory populations (EPG / EPGt / PEN_a / PEN_b / PEG) is not resolved by the present analysis, and the same identifiability gap should be expected there until ruled out.

Identifiable computation, non-identifiable implementation. Three of the four parameterisations (Known-ODE RNN, fully connected RNN, GNN) reach essentially identical heading accuracy on path integration but realise the computation through qualitatively different machinery: different per-edge weights (Sect. 3), different presence or absence of off-block edges (Known-ODE vs. fully connected), and different per-neuron intrinsic functions (RNN’s hardcoded $-h/\tau, \sigma$ vs. GNN’s learned f_θ, g_ϕ). The fourth, frozen- W^{rec} control fails outright (Fig. 5), ruling out the simplest reading that the raw connectome alone instantiates the algorithm. What remains *identifiable* across the three that converge? (i) The functional architecture: all three implement a ring-attractor with a bump that propagates as heading changes, visible as a diagonal bump migration in Fig. 12. (ii) The input–output routing: in all three trained networks, the EPG / PEN / $\Delta 7$ neurons participate in the bump while ER6 / PEG cells do not — a consequence of the $\hat{W}^{\text{in}}$ row structure they share (stimulus routed to PEN sub-populations only, initial-heading impulse free for all). What is *not* identifiable: per-edge $\hat{W}^{\text{rec}}$ (Sect. 3), the form of the per-neuron drift, the form of the firing-rate non-linearity, and the anatomical-neuron $\rightarrow$ preferred-phase mapping (different trained networks assign different anatomical neurons to the same phase φ along the ring). Optimising the path-integration input–output map fixes the ring-attractor computation; the implementation remains under-determined at every level we probe.

Biologically aligned vs alternative cell-type allocations. The three models reach statistically equivalent heading precision ($I_{\text{pop}} = 3.43/3.05/3.35$ bits; per-step posterior $\sigma_{\theta|\hat{h}} \sim 10^\circ$) but parcellate that information across cell types in mutually inconsistent ways (Fig. 13, Tab. 1). The Known-ODE RNN is the most biologically aligned: EPG / EPGt and $\Delta 7$ carry the dominant HD information while ER6 remains essentially tonic, matching the canonical CX picture (Seelig and Jayaraman, 2015; Kim et al., 2017; Green et al., 2017) in which EPG neurons encode heading and $\Delta 7$ provides the

recurrent ring-inhibitory scaffold. The fully connected RNN saturates EPGt and shifts the per-neuron champion to $\Delta 7$; the GNN goes further and implements a $\Delta 7 \leftrightarrow \text{ER6}$ swap that nonetheless preserves the ring-attractor computation. The biological division of labour is therefore not forced by the path-integration task alone: the same input–output map admits alternative circuit implementations that reallocate HD information away from the canonical $\Delta 7$ -centred inhibitory pathway.

Relation to Hulse et al. 2026. The close companion to our analysis is the *hidden-symmetries* framework of Hulse et al. (2026), which shows how ring-attractor dynamics can be preserved under heterogeneous connectivity through hidden symmetries embedded in cloned network architectures. Their study combines analytically constructed symmetry-preserving networks with unconstrained optimisation results, arguing that the CX attractor is robust to substantial microscopic heterogeneity. Our analysis asks a complementary question: given successful behavioural recovery on the same path-integration task, which aspects of the recovered implementation are actually identifiable? Our per- $(i, \alpha, \text{instance})$ null directions (Sect. 3) overlap closely with the clone structure highlighted in their framework, but we further show that the degeneracy extends beyond connectivity to the intrinsic neuronal dynamics themselves.

(i) Per-cell-type perturbations are locally null but do not remain invariant under full collapse. Mixing clones across PB glomeruli (the per- (i, α) partition) breaks the readout at $r = 0.36$. Per-cell-type groupings are not identified as clones by Hulse et al. (2026) — their analysis works at the per-glomerulus instance level — and our empirical test confirms it: lumping same-type neurons together is too coarse to saturate the null space. **(ii) Per-unit clones more closely track the invariant subspace.** Our per- $(i, \alpha, \text{instance})$ partition (Sect. 3) groups the same-type, same-PB-glomerulus neurons that Hulse et al. identify as clones, and sum-preserving collapses inside each group leave the rollout at $r = 0.83$ even after zeroing 50.2% of the edges — the structural symmetry is operationally real. The natural unit of redundancy is the glomerulus, not the cell type, exactly as the symmetry framework predicts. **(iii) The connectome’s wiring is necessary but not sufficient.** The frozen- W^{rec} control (Fig. 5) fails to integrate, so the symmetry classes of the raw connectome *cannot* on their own instantiate the attractor: learned per-edge magnitudes inside those classes are doing real work. This is the complementary statement to Hulse et al. (2026): their robustness proof assumes a working attractor; we show that training, not topology, supplies it. **(iv) The functional form is also non-unique.** The hidden-symmetries analysis fixes a sigmoidal σ and a linear leak as part of the model; our GNN result (Fig. 7) shows that behaviourally indistinguishable dynamics arise from a learned (f_θ, g_ϕ) pair that *neither* reduces to $-h/\tau$ nor to the canonical sigmoid. The symmetry classes therefore explain one layer of degeneracy — hidden symmetries within a fixed neuronal model — whereas our GNN result adds a second layer: the neuronal dynamics themselves are also non-identifiable. The per-cell-type HD MI analysis (Fig. 13, Tab. 1) sharpens this point: alternative signalling functions (f_θ, g_ϕ) do recover the behaviour, but they do so by routing HD information through cell types that disagree with the biological prior (EPGt silent in the fully connected RNN, $\Delta 7$ silent in the GNN, while in calcium imaging (Seelig and Jayaraman, 2015; Kim et al., 2017; Green et al., 2017) EPG, EPGt, PEN, PEG, and $\Delta 7$ are all HD-tuned). Other signalling functions therefore recover the function but not the biological cell-type assignment.

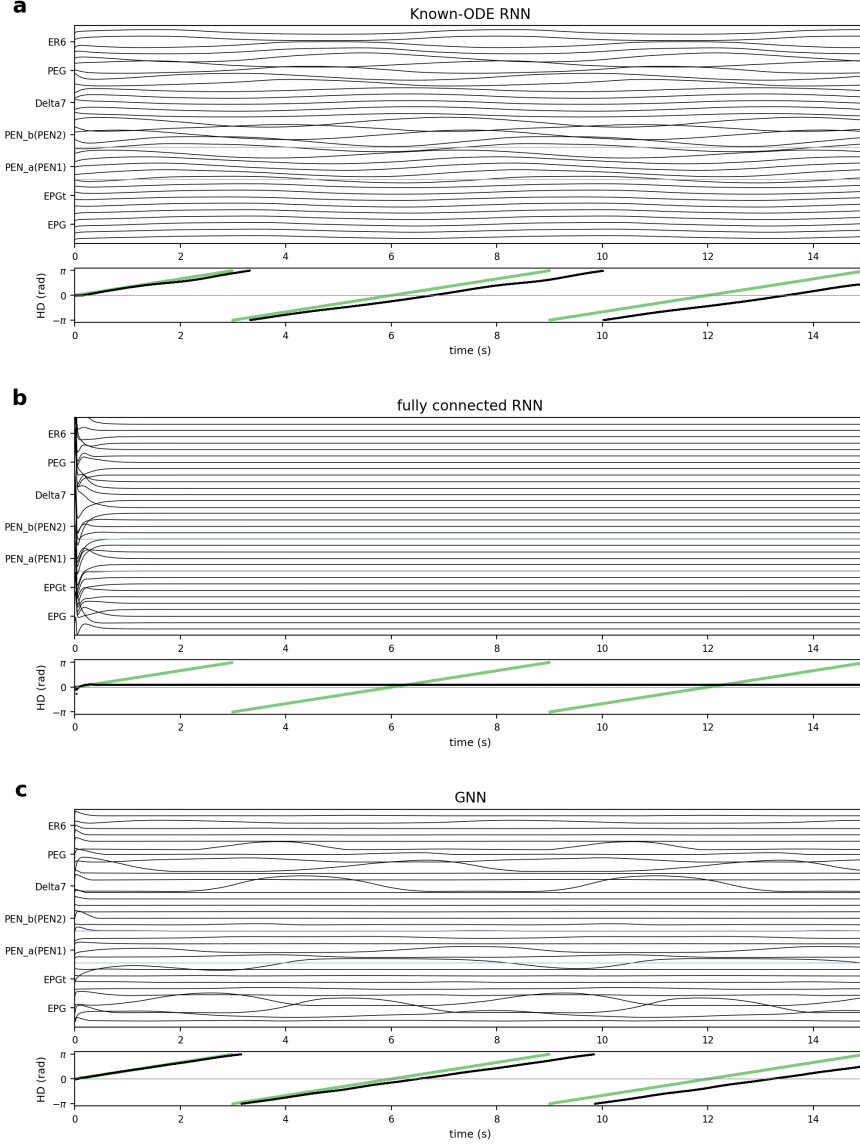


Figure 9: **Voltage traces of the three converged models (Known-ODE RNN, fully connected RNN, GNN; the frozen- W^{rec} control fails and is omitted) under a deterministic constant angular velocity stimulus $\omega = 60^\circ/\text{s}$ ($T = 1\,500$ frames at $dt = 0.01\text{ s} = 15\text{ s} = 2.5$ full turns).** Same per-model trace layout as Fig. 10: 5 representative neurons per cell type stacked vertically, with the L / R PEN stim overlay (light red / light blue) on the first PEN_a / PEN_b row of each cluster; bottom strip shows wrapped HD GT in green vs. decoded in black. Under a constant- ω drive the head direction sweeps linearly and the bump migration is periodic, so each EPG / PEN / $\Delta 7$ neuron rings up and down at a fixed phase offset — the canonical ring-attractor signature. **(a)** Known-ODE RNN. **(b)** Fully connected RNN. **(c)** GNN. All three reproduce the constant-rate HD sweep with periodic bump migrations.

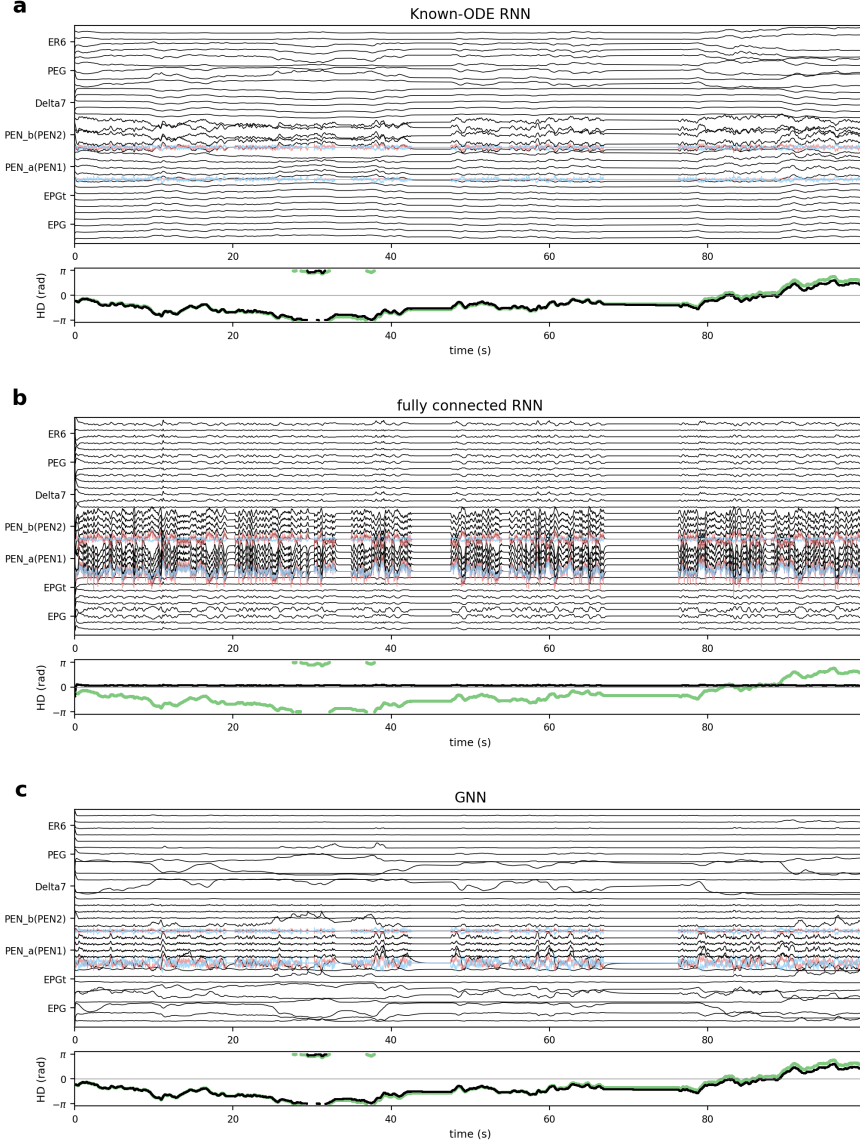


Figure 10: **Voltage traces of the three converged models (Known-ODE RNN, fully connected RNN, GNN; the frozen- W^{rec} control fails and is omitted) under the natural OU velocity stimulus.** 2×2 panel: each tile is the same task-traces layout (5 representative neurons per cell type stacked vertically; PEN_a / PEN_b clusters carry the L / R stim overlay in light red / blue; bottom strip shows wrapped HD GT in green, decoded in black). **(a)** Known-ODE RNN (connectome topology and Dale signs imposed). **(b)** fully connected RNN (only the column-wise Dale sign of each presynaptic neuron is enforced — topology mask dropped). **(c)** GNN (Sect. 2; positive-monotonicity prior $\partial g_\phi / \partial \hat{h} \geq 0$ enforced). All three reproduce path integration on this rollout (decoded HD tracks GT) but the underlying subthreshold dynamics differ visibly: the two RNN variants drive every neuron through a full bump-traversal cycle as heading changes, while the GNN concentrates most cells in a near-zero silent plateau and lets only a small bump-encoding sub-population carry the integration (cf. Fig. 4).

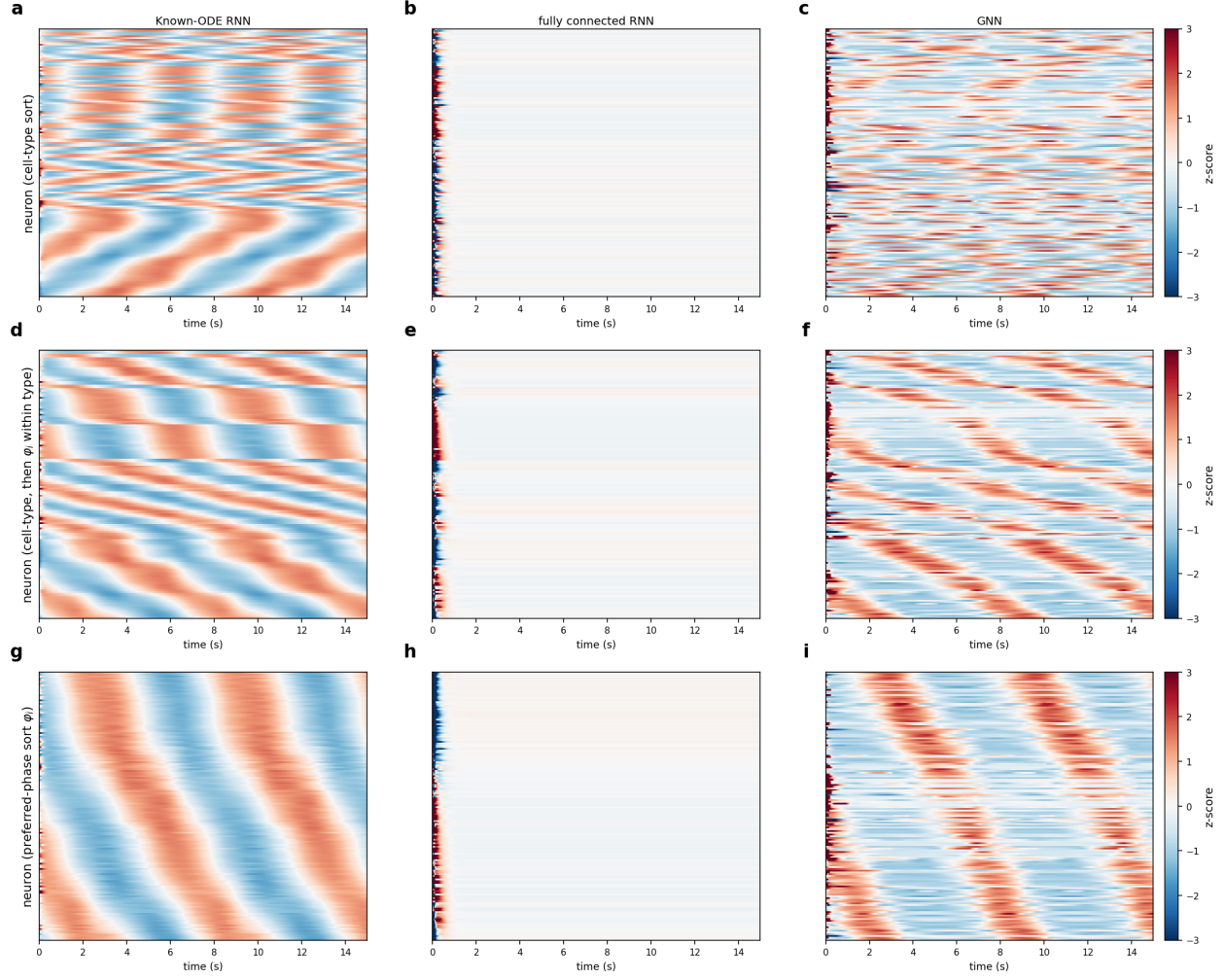


Figure 11: **All-neurons-by-time kinographs of the four models' subthreshold state under a deterministic constant angular velocity stimulus $\omega = 60^\circ/\text{s}$ ($T = 1500$ frames at $dt = 0.01$ s = 15 s = 2.5 full turns), in three row orderings.** Each tile shows the per-neuron z-scored $(\hat{h}_i(t) - \langle \hat{h}_i \rangle) / \text{std}(\hat{h}_i)$ for all 156 CX neurons; colour fixed at ± 3 z-scores so the columns are directly comparable regardless of each model's absolute voltage scale. Columns: Known-ODE RNN, fully connected RNN, GNN. **Row 1 (a–c, cell-type sort, ER6 $\rightarrow$ EPG top-to-bottom).** Same row ordering as Fig. 10. The Known-ODE RNN (a) shows a bump migrating through the EPG / PEN / $\Delta 7$ blocks as heading changes; the fully connected RNN (b) shows the same pattern with additional weak background activity in the formerly-zero off-block entries; the GNN (c) retains the bump-encoding sub-population in EPG / PEN / $\Delta 7$ but the cell-type ordering scrambles the bump migration into apparent noise, because same-type cells encode different glomeruli. **Row 2 (d–f, cell-type primary, preferred-phase secondary within type).** Cell-type blocks preserved (ER6 $\rightarrow$ EPG), but rows within each type are sorted by their preferred heading phase $\varphi_i = \arg(\sum_t (\hat{h}_i(t) - \langle \hat{h}_i \rangle) e^{i\theta(t)})$. Each block now reads as a small ring: the per-block diagonals all share the same instantaneous phase, making the ring-attractor bump visible without losing the anatomical type labels. **Row 3 (g–i, pure preferred-phase sort, low- φ at top).** Under this ordering the three converged models reveal the same diagonal bump-migration signature — the canonical ring-attractor readout. The constant- ω drive fixes the bump-migration slope to ω , so the phase-sorted diagonals are clean and parallel. All three resolve the path-integration task with a ring-attractor structure built on *different* mappings from anatomical neuron-index to φ .

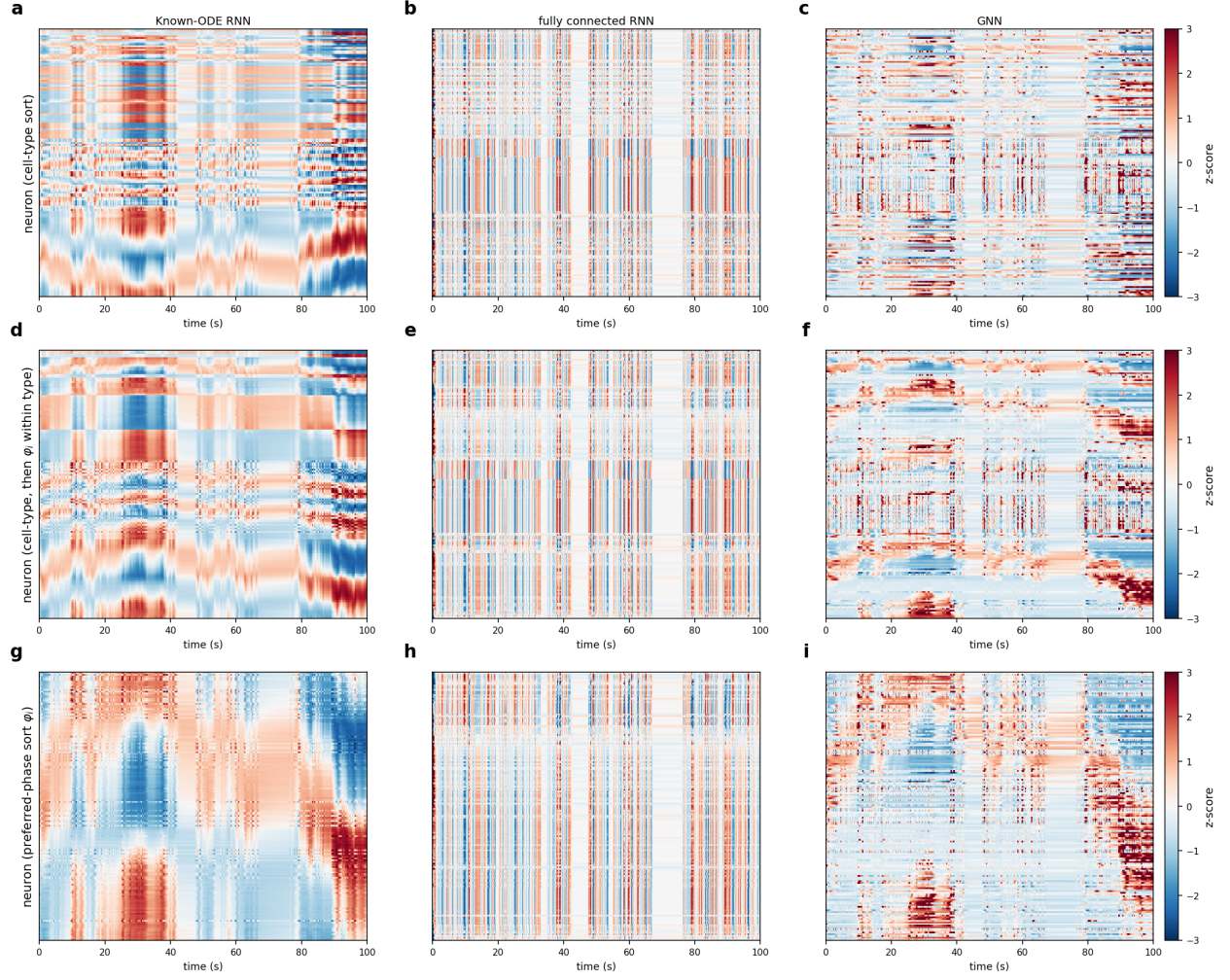


Figure 12: **Same row orderings as Fig. 11, but under the natural OU velocity stimulus ($T = 10\,000$ frames ≈ 100 s).** 3-row layout: cell-type sort (a–d), cell-type primary with φ_i secondary within type (e–h), and pure preferred-phase sort (i–l). Replacing the deterministic constant- ω drive of Fig. 11 with the stochastic OU drive used for training preserves the same ring-attractor signature in all three converged models. The phase-sorted diagonals (i–l) now wander with the OU heading trajectory and pause whenever the stimulus inserts a standing interval, but the underlying ring-attractor structure is identical, confirming that the readout is not an artefact of the deterministic drive.

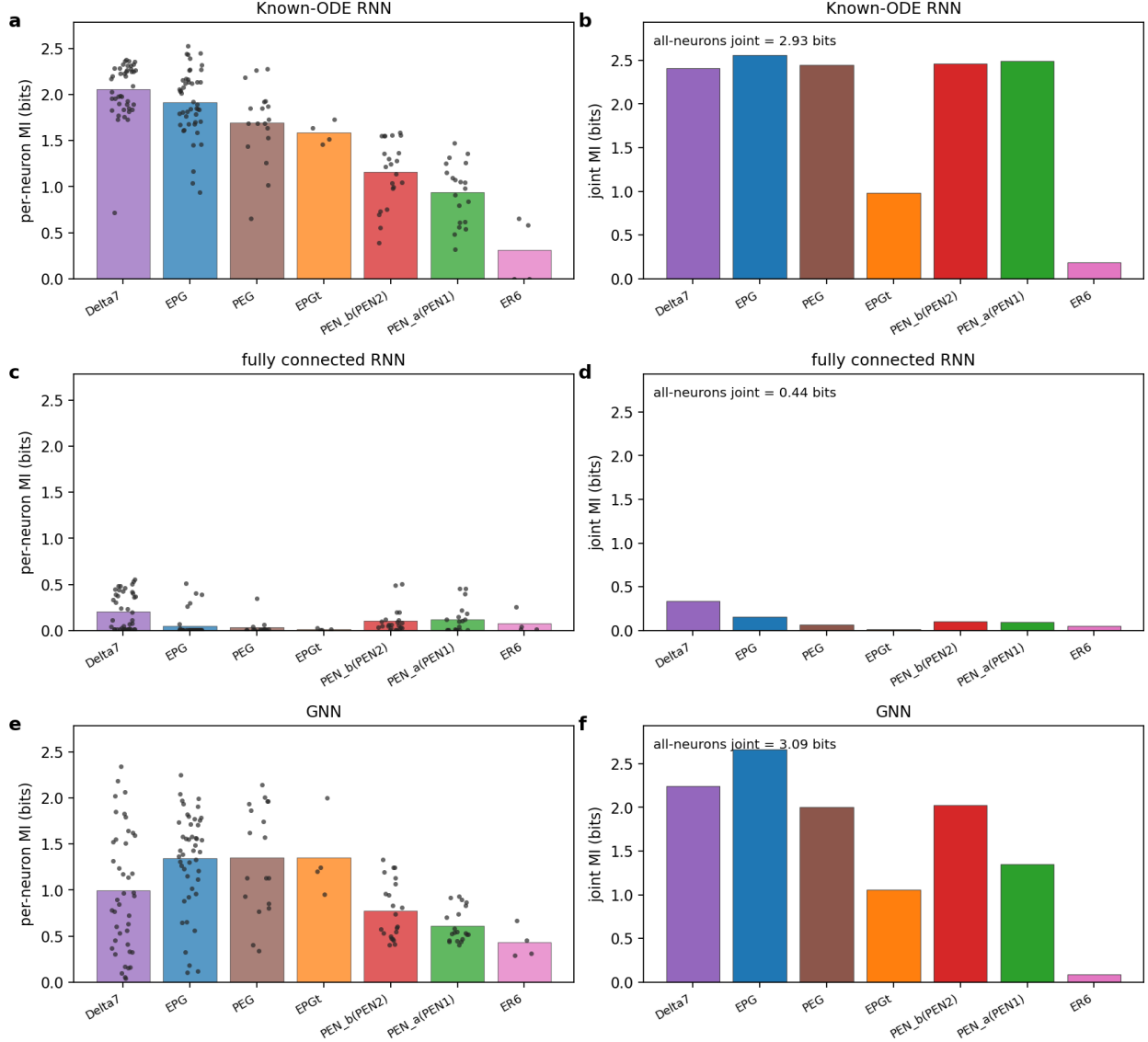


Figure 13: **Per-cell-type head-direction mutual information for the three converged CX models**, evaluated along the natural OU velocity rollout ($T = 10000$ frames, $n_\theta = 64$ angular bins). One row per model. **Left (a, c, e)** per-neuron MI $I(\sigma(\hat{h}_i); \theta)$ in bits (bars = per-cell-type mean, dots = individual neurons); **right (b, d, f)** joint per-type MI $I(\{\sigma(\hat{h}_i)\}_{i \in t}; \theta)$ (CV-logreg lower bound), with the population joint across all 156 neurons annotated inline. Cell types are ordered by the Known-ODE per-neuron mean so vertical comparison reads as a per-cell-type fingerprint. The three models reach matched behavioural precision: population joint MI 3.43/3.05/3.35 bits for Known-ODE / FC / GNN, predicting a Gaussian-channel per-step posterior $\sigma_{\theta|\hat{h}} = \sigma_\theta 2^{-I_{\text{pop}}} \approx 9.6/12.5/10.2^\circ$ ($\sigma_\theta = \pi/\sqrt{3} \approx 1.81$ rad), of the same order as the directly measured circular RMSEs ($2.5/12.4/37.0^\circ$ constant- ω ; $25.8/28.9/6.1^\circ$ OU; rollout error compounds integration drift on top of the per-step decoding floor). Despite matched precision, HD coding is distributed across cell types in mutually inconsistent ways: ER6 silent in the Known-ODE RNN, EPGt silent in the FC, $\Delta 7$ silent in the GNN — the per-cell-type fingerprint that motivates the alternative cell-type allocations discussed in the per-model Results paragraphs.

Table 1: **Per-cell-type HD MI summary across the three converged models**, from the same OU rollout as Fig. 13 ($n_\theta = 64$ bins). $\bar{I}_i$ is the per-neuron mean $I(\sigma(\hat{h}_i); \theta)$ within the type; I_{joint} is the population MI of the whole type (CV-logreg lower bound). Redundancy $\rho = \sum_i I_i / I_{\text{joint}}$: large ρ means the cells of that type share near-identical HD tuning; $\rho \rightarrow \infty$ marks a silent population. ER6 is silent only in the Known-ODE RNN; EPGt is silent only in the fully connected RNN; $\Delta 7$ is silent only in the GNN.

type	n	Known-ODE RNN			fully connected RNN			GNN		
		$\bar{I}_i$	I_{joint}	ρ	$\bar{I}_i$	I_{joint}	ρ	$\bar{I}_i$	I_{joint}	ρ
EPG	46	1.21	3.04	18.3	0.48	2.24	9.9	1.19	2.69	20.4
EPGt	4	1.54	1.64	3.8	0.01	0.005	8.8	1.14	1.25	3.6
PEN _a	20	0.78	2.56	6.1	0.49	1.94	5.0	0.98	2.56	7.6
PEN _b	22	0.88	2.74	7.0	0.42	1.91	4.8	1.41	2.84	10.9
$\Delta 7$	42	0.95	2.79	14.3	1.15	2.82	17.1	0.11	0.15	30.1
PEG	18	1.12	2.66	7.6	0.73	2.12	6.2	0.81	1.71	8.5
ER6	4	0.001	0.000	∞	0.96	1.68	2.3	1.18	1.13	4.2
all (156)	–	$I_{\text{pop}} = 3.43$ bits			$I_{\text{pop}} = 3.05$ bits			$I_{\text{pop}} = 3.35$ bits		

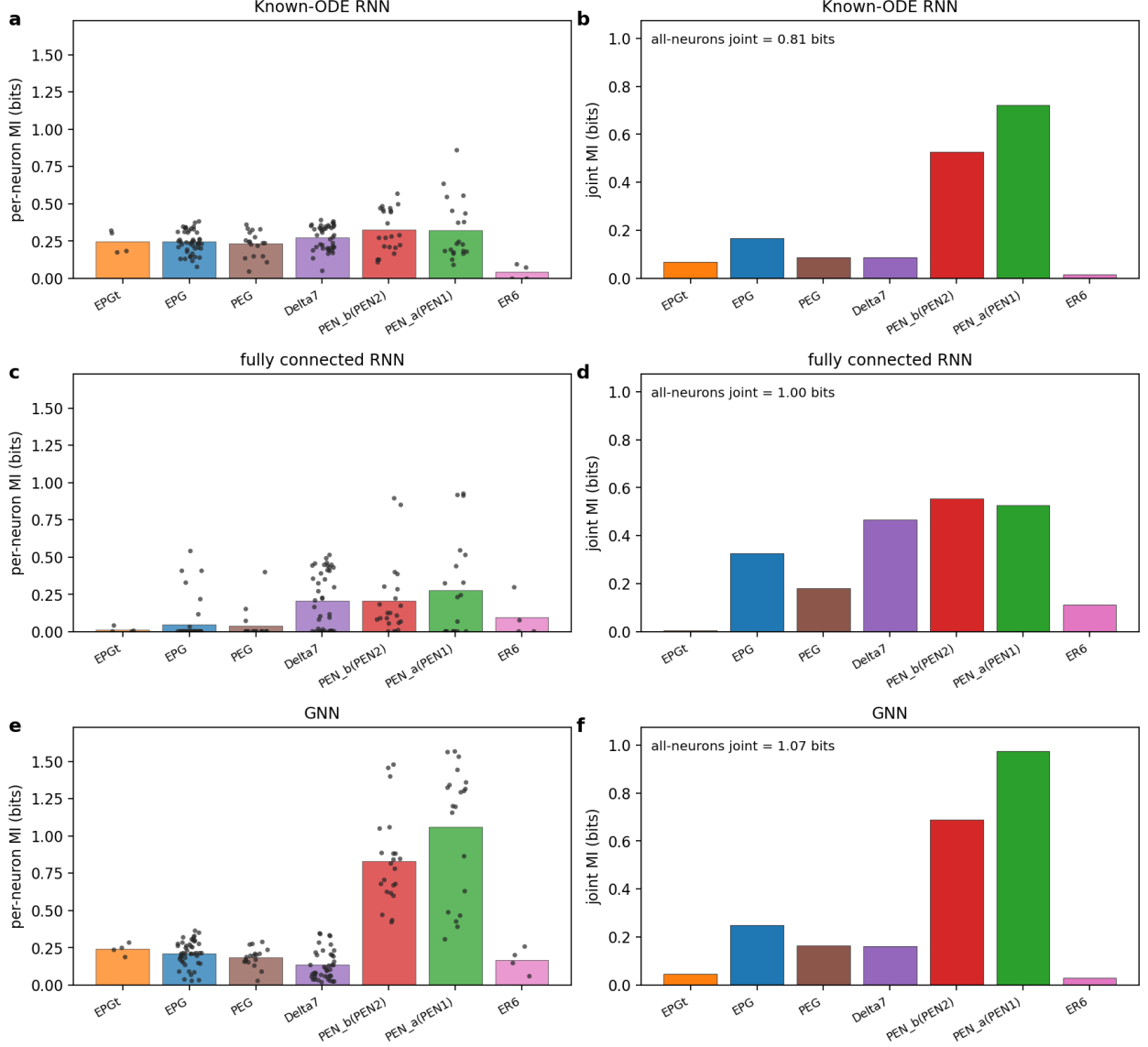


Figure 14: Per-cell-type angular-velocity mutual information for the three converged CX models. Same OU rollout, layout, and cell-type ordering as Fig. 13 with θ replaced by the instantaneous angular velocity ω ($n_\omega = 64$ uniform bins over the 99th-percentile $|\omega|$ range, $\sigma_\omega \approx 37.5^\circ/\text{s}$). Rows: Known-ODE / FC / GNN; left: per-neuron $I(\sigma(\hat{h}_i); \omega)$; right: joint per-type MI with the population joint annotated inline. The PEN_a and PEN_b sub-populations are the top ω -encoders in all three models ($\bar{I}_{\text{PEN}_a/\text{PEN}_b} = 0.24/0.19, 0.14/0.15, 0.20/0.23$ bits per neuron for Known-ODE / FC / GNN), a structural consequence of the four-scalar PEN velocity gate (Eq. (4)). Absolute ω MI is an order of magnitude below the HD MI because ω is a short-timescale derivative of the integrated heading. Population joint MI is 0.88/0.84/0.49 bits, giving Gaussian-channel per-step posteriors $\sigma_{\omega|\hat{h}} \approx \sigma_\omega 2^{-I_{\text{pop}}} = 20/21/27^\circ/\text{s}$ — of the same order as the empirical prior, so a single time step barely constrains ω . Path integration therefore relies on accumulating this noisy per-step estimate into the much tighter $10\text{--}12^\circ$ heading posterior of Fig. 13. The GNN’s reduced ω MI is consistent with its tight-bump strategy (Figs. 4 and 10): once ω has been integrated into θ , the per-step velocity signal is discarded by most of the network. $\Delta 7$, the second-strongest ω -encoder in Known-ODE / FC, is also silent here in the GNN — a downstream echo of the $\Delta 7 \leftrightarrow \text{ER6}$ swap in Fig. 13.

Table 2: **Per-cell-type angular-velocity MI summary**, same rollout and definitions as Tab. 1 but with ω as the target ($n_\omega = 64$ uniform bins, range $[-q_{99}, q_{99}]$). PEN_a and PEN_b are the dominant ω -encoders in all three models. The Gaussian-channel per-step posterior s.d. $\sigma_{\omega|\hat{h}}$ is computed from the population joint MI as $\sigma_\omega 2^{-I_{\text{pop}}}$, with $\sigma_\omega \approx 37.5^\circ/\text{s}$.

type	n	Known-ODE RNN			fully connected RNN			GNN		
		$\bar{I}_i$	I_{joint}	ρ	$\bar{I}_i$	I_{joint}	ρ	$\bar{I}_i$	I_{joint}	ρ
EPG	46	0.20	0.38	24.0	0.07	0.15	19.8	0.15	0.16	42.6
EPGt	4	0.21	0.02	36.0	0.01	0.000	–	0.15	0.05	12.9
PEN_a	20	0.24	0.60	8.1	0.14	0.43	6.4	0.20	0.30	13.3
PEN_b	22	0.19	0.57	7.4	0.15	0.62	5.2	0.23	0.31	16.3
$\Delta 7$	42	0.16	0.29	22.9	0.18	0.34	21.8	0.03	0.01	91.0
PEG	18	0.17	0.27	11.2	0.12	0.07	28.7	0.12	0.05	44.1
ER6	4	0.001	0.000	–	0.16	0.03	24.9	0.12	0.04	12.3
all (156)	–	$I_{\text{pop}} = 0.88; \sigma_{\omega \hat{h}} = 20.4^\circ/\text{s}$			$I_{\text{pop}} = 0.84; 21.0^\circ/\text{s}$			$I_{\text{pop}} = 0.49; 26.6^\circ/\text{s}$		

Appendix A: Degeneracy of the inverse problem

Above, we demonstrated that the trained recurrent neural network with imposed connectome topology and Dale signs reproduces the path-integration task at near-perfect accuracy across 10 independently trained CV folds (Fig. 3, Fig. 8). Here, we ask how degenerate the inverse problem is: how many distinct recurrent weight matrices $\hat{W}^{\text{rec}}$ reproduce the same voltage trajectories, and which directions in $\hat{W}^{\text{rec}}$ -space are actually invariant for the trained solutions? All numerical results in this appendix are reported as mean $\pm$ std across the same 10 CV folds used in Fig. 8.

We fix the values of the bias $\hat{b}$, the time constant τ , and the input weights $\hat{W}^{\text{in}}$ at their trained values, and rearrange Eq. (1) to isolate the incoming synaptic weights:

$$\sum_{j \in \mathcal{N}_i} \hat{W}_{ij}^{\text{rec}} \sigma(\hat{h}_j(t)) = \tau \frac{d\hat{h}_i(t)}{dt} + \hat{h}_i(t) - \hat{b}_i - I_i(t), \quad (14)$$

where $I_i(t) = \sum_l \hat{W}_{il}^{\text{in}} u_l(t)$ is the per-neuron projected input from Eq. (4). Defining the presynaptic gated activity $h_j(t) = \sigma(\hat{h}_j(t))$ and the residual $b_i(t) = \tau d\hat{h}_i(t)/dt + \hat{h}_i(t) - \hat{b}_i - I_i(t)$, and stacking the T time-point constraints into matrix form, yields an independent linear system for each postsynaptic neuron i :

$$\mathbf{H}_i \mathbf{w}_i = \mathbf{b}_i, \quad (15)$$

where $\mathbf{H}_i \in \mathbb{R}^{T \times d_i}$ is the presynaptic activity matrix, d_i is the in-degree of neuron i , $\mathbf{w}_i \in \mathbb{R}^{d_i}$ the incoming recurrent weight vector, and $\mathbf{b}_i \in \mathbb{R}^T$ the stacked residuals. This holds for each postsynaptic neuron $i = 1, \dots, N$, where $N = 156$ and d_i ranges from 1 to 98. The null space $\ker(\mathbf{H}_i) = \{\boldsymbol{\delta} \in \mathbb{R}^{d_i} : \mathbf{H}_i \boldsymbol{\delta} = \mathbf{0}\}$ characterises the degeneracy: any $\mathbf{w}_i + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \in \ker(\mathbf{H}_i)$ satisfies Eq. (15) equally and produces identical dynamics. In the following, we characterise the dynamical equivalence class of $\hat{W}^{\text{rec}}$, the set of all recurrent weight matrices that reproduce the trained model’s voltage trajectories.

Two SVD-based bounds on the null space (inconclusive). We solve Eq. (15) per neuron via column-equilibrated normal equations and the eigendecomposition of $\mathbf{H}_i^\top \mathbf{H}_i$, and quantify $\dim \ker(\mathbf{H})$ in two complementary ways. *Strict per-neuron SVD*: the eigendecomposition flags directions as *null* (relative eigenvalue $\leq T \cdot 10^{-22}$) or *sloppy* ($\leq T \cdot 10^{-12}$); summed over i this counts only 6.3% of edges (0 null + 650 sloppy). *Global-SVD coarse bound*: by rank-nullity, $d_i = \text{rank}(\mathbf{H}_i) + \dim \ker(\mathbf{H}_i)$. Each per-neuron activity matrix $\mathbf{H}_i$ is a column-submatrix of the global activity matrix $\mathbf{H} \in \mathbb{R}^{T \times N}$ stacking sigmoid gated activities $\sigma(\hat{h}_j(t))$ of all 156 neurons over $T = 8,000$ subsampled frames; therefore $\text{rank}(\mathbf{H}_i) \leq r$, where r is the effective rank of $\mathbf{H}$ at a chosen variance threshold, and

$$\dim \ker(\mathbf{H}) \approx \sum_{i=1}^N \max(0, d_i - r). \quad (16)$$

At 99% variance, $r = 6$ — only six modes carry 99% of the population variance, reflecting the low-dimensional ring-attractor manifold — and the bound spans 9,327 edges (90.9%). Tightening the threshold to 99.9% raises r to 15 and shrinks the bound to 7,923 edges (77.2%); the null space remains large at every threshold. The two estimates disagree by more than an order of magnitude and neither is conclusive on its own. To pin down which directions are actually invariant for the trained model, we apply controlled perturbations to specific groups of edges and roll out the network. We test two partition schemes in the next two paragraphs.

Per cell type. For each postsynaptic neuron i and presynaptic cell type α , the $k_{i,\alpha}$ same-type incoming edges form a candidate (i, α) group; pairs with $k_{i,\alpha} \geq 2$ account for 9,400 edges (91.6% of the 10,263-edge connectome support; structural, identical across folds). The sum-zero perturbation is the natural test for this partition: when the $k_{i,\alpha}$ same-type presynaptic neurons carry correlated subthreshold activity, the recurrent drive $\sum_m W_{i,j_m} \sigma(\hat{h}_{j_m}(t)) \approx \sigma(\hat{h}(t)) \sum_m W_{i,j_m}$ depends only on the group sum, so any perturbation on the same-type weights $W_{i,j_m} \rightarrow W_{i,j_m} + \delta_{j_m}$ with $\sum_m \delta_{j_m} = 0$ leaves the dynamics invariant under the lockstep assumption (Allier et al., 2026a). We test this candidate null direction empirically by perturbing each fold’s groups by a sum-zero vector at amplitude $\lambda \cdot \text{mean}(|W_{\text{group}}|)$ and rolling out for 1,000 frames against that fold’s own GT voltage; reported values average over the 10 folds (Tab. 3). At $\lambda = 1$, all seven cell types preserve the dynamics ($R_{\mathbf{W}}^2 \in [0.97, 1.00]$ across folds, Pearson r in $[0.96, 1.00]$): the predicted directions are real *locally* around the trained $\hat{W}^{\text{rec}}$, and the result is reproducible across folds (per-cell-type r SDs ≤ 0.09). At $\lambda = 2$ all r remain above 0.88 (PEG and $\Delta 7$ the loosest, the latter with the largest inter-fold spread, $r = 0.88 \pm 0.14$); at $\lambda = 4$, $R_{\mathbf{W}}^2$ drops to $[0.54, 0.97]$ and $\Delta 7$ /PEG rollouts deteriorate most strongly ($r = 0.83 \pm 0.08$ / 0.86 ± 0.12). Pushing all the way to the extreme by collapsing each (i, α) group of $k_{i,\alpha}$ edges onto a single representative edge $e^* = \arg \max_e |W_e|$ — setting $W_{e^*} = \sum_{e \in \text{group}} W_e$ (sum-preserving) or W_{e^*} chosen by a one-scalar least-squares fit that minimises the per-group dynamics residual (calibrated), and zeroing the other $k_{i,\alpha} - 1$ edges — removes 91.6% of the edges. This is the analogue of the operation that preserves dynamics at $r \approx 0.95$ on the visual-system biomodel of (Allier et al., 2026a), but it breaks the CX heading readout for both variants: sum-preserving rollout $r = 0.27 \pm 0.24$, calibrated rollout $r = 0.13 \pm 0.18$ (both averaged over 10 folds). The per-cell-type null space is therefore approximately null around the trained weights but cannot be saturated by a single representative per cell-type pair.

Per unit. The per- (i, α) partition lumps same-type neurons regardless of which PB glomerulus they innervate, but in the CX two EPGs in different glomeruli encode different headings and are not clones. The natural refinement groups same-type neurons by their hemibrain instance, the per-cell annotation that identifies a neuron’s anatomical computational unit: for EPG/EPGt/PEG/PEN it is the PB glomerulus (e.g., *EPG(PB08)_L3* marks an EPG neuron in glomerulus L3); for $\Delta 7$ it is the set of glomeruli the neuron spans (e.g., *Delta7(PB15)_L1L9R8_R*); for ER6 it is the hemisphere. Same-type neurons sharing an instance are clones as discussed in Hulse et al. (2026). The $(i, \alpha, \text{instance})$ partition retains 5,150 candidate edges (50.2% of 10,263; structural, identical across folds). Applying the same collapse defined above to each $(i, \alpha, \text{instance})$ group — zeroing 50.2% of the edges instead of 91.6% — yields, across 10 folds, sum-preserving rollout $r = 0.55 \pm 0.20$ and calibrated rollout $r = 0.25 \pm 0.28$, against $r = 0.27 \pm 0.24$ / 0.13 ± 0.18 for the analogous operations under the per- (i, α) partition. The per-unit sum-preserving collapse therefore continues to track heading more robustly than the per-cell-type one, confirming that the finer (instance-level) partition more closely respects the trained model’s null space. The single-fold $r = 0.98$ calibrated value reported in the predecessor (no-TV-prior) analysis no longer holds once $\hat{W}^{\text{rec}}$ is biased away from the per-instance null directions by the TV regulariser used in the present training recipe (Sect. 2): across folds, the calibrated collapse falls below the sum-preserving one and has large inter-fold spread (0.25 ± 0.28 , 95% CI spans the full $[0, 0.8]$ range). Mechanistically, this says that the trained $\hat{W}^{\text{rec}}$ is not symmetric within an $(\alpha, \text{instance})$ clone group: each fold finds a structurally similar but non-interchangeable per-clone allocation, and a single-scalar OLS fit cannot reproduce that asymmetry as accurately as the structural sum. In a retinotopic visual system (Allier et al., 2026a), the (i, α) partition already saturates because columns of the same type encode the same scene; the CX requires the finer $(i, \alpha, \text{instance})$ partition because heading is encoded per-glomerulus,

not per-type — but on the trained-weight side, the calibrated saturation that holds for the visual system is broken here by the TV-shaped solutions.

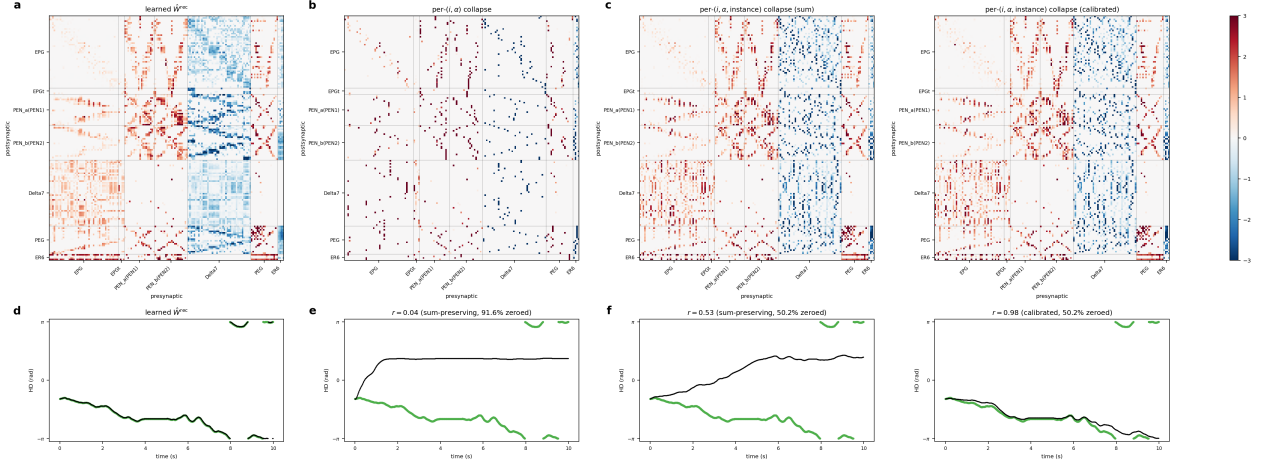


Figure 15: **Sparsifying $\hat{W}^{\text{rec}}$ along the candidate null directions.** Top row (a-d): recurrent matrices, sorted by cell type (block boundaries marked) and z-scored on the same scale (computed from the GT non-zero entries, clipped to ± 3). Bottom row (e-h): heading on one OU test trial decoded from each rollout; true HD in light green, decoded HD in black, both wrapped to $(-\pi, \pi]$. **(a, e)** Learned $\hat{W}^{\text{rec}}$ from the trained CxTaskRNN: GT and decoded HD coincide. **(b, f)** Per- (i, α) sum-preserving collapse, 91.6% of edges zeroed: the bump-coding blocks (EPG, PEN_a , PEN_b , $\Delta 7$) lose most of their structure and the decoded HD drifts away from GT ($r = 0.04$). **(c, g)** Per- $(i, \alpha, \text{instance})$ sum-preserving collapse, 50.2% of edges zeroed: the block structure is preserved and the decoded HD partially tracks GT ($r = 0.53$). **(d, h)** Same per- $(i, \alpha, \text{instance})$ partition but with a *calibrated* one-scalar-per-group fit (each group of k same-instance clones is replaced by a single representative edge whose weight is set by least-squares to minimise the per-group dynamics residual): the same 50.2% of edges are zeroed yet the bump tracks GT at $r = 0.98$, demonstrating that the per- $(i, \alpha, \text{instance})$ partition is near-fully saturable when intra-unit clones are summarised by a calibrated scalar instead of a naïve sum.

Cell Type	$\lambda = 1$		$\lambda = 2$		$\lambda = 4$	
	$R_{\mathbf{W}}^2$	r	$R_{\mathbf{W}}^2$	r	$R_{\mathbf{W}}^2$	r
EPG	0.998 ± 0.000	0.999 ± 0.002	0.992 ± 0.002	0.997 ± 0.003	0.970 ± 0.007	0.975 ± 0.034
EPGt	0.998 ± 0.001	0.990 ± 0.029	0.992 ± 0.003	0.953 ± 0.095	0.969 ± 0.012	0.949 ± 0.115
PEN_a(PEN1)	0.994 ± 0.001	0.986 ± 0.028	0.974 ± 0.005	0.991 ± 0.013	0.898 ± 0.022	0.936 ± 0.066
PEN_b(PEN2)	0.988 ± 0.003	0.964 ± 0.088	0.951 ± 0.012	0.981 ± 0.016	0.803 ± 0.050	0.940 ± 0.088
Delta7	0.971 ± 0.004	0.960 ± 0.049	0.886 ± 0.015	0.876 ± 0.142	0.543 ± 0.059	0.833 ± 0.078
PEG	0.989 ± 0.003	0.992 ± 0.006	0.957 ± 0.012	0.979 ± 0.017	0.826 ± 0.050	0.859 ± 0.117
ER6	0.991 ± 0.002	0.999 ± 0.002	0.963 ± 0.006	0.990 ± 0.020	0.851 ± 0.025	0.985 ± 0.028

Table 3: **Single-type sum-zero variants across perturbation amplitudes λ , mean $\pm$ std over 10 CV folds.** For each cell type with $k_{i,\alpha} \geq 2$ groups, all degenerate weights are perturbed by a sum-zero vector at amplitude $\lambda \cdot \text{mean}(|W_{\text{group}}|)$ and rolled out for 1,000 frames against that fold’s own GT voltage trajectory. Columns: connectivity $R_{\mathbf{W}}^2$ (between the perturbed and trained recurrent matrix on the connectome support) and rollout Pearson r at $\lambda \in \{1, 2, 4\}$, averaged across the 10 CV folds with the inter-fold standard deviation as $\pm$ band. At $\lambda = 1$ every type is preserved with $r \geq 0.96 \pm 0.09$; both $R_{\mathbf{W}}^2$ and r decrease with λ , confirming that the candidate null directions are only approximately null at large perturbations, and the inter-fold spread grows monotonically with λ .

References

- C. Allier et al. Graph neural networks recover interpretable circuit models from neural activity. In preparation (NeurIPS 2026 submission), 2026.
- C. Allier and S. Saalfeld. A four-MLP GNN framework with explicit temporal filtering, and the cost-structured route to emergent circuit architecture. Internal technical report `GNN_Transformer.tex`, 2026.
- J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl. Neural message passing for quantum chemistry. In *ICML*, 2017.
- B. K. Hulse, P. B. Aneesh, S. Romani, V. Jayaraman, and A. M. Hermundstad. How hidden symmetries in network connectivity support robust attractor dynamics in the fly’s neural compass. *bioRxiv*, 2026. doi:10.64898/2026.05.18.725766. URL <https://www.biorxiv.org/content/10.64898/2026.05.18.725766v1>.
- J. Green, A. Adachi, K. K. Shah, J. D. Hirokawa, P. S. Magani, and G. Maimon. A neural circuit architecture for angular integration in *Drosophila*. *Nature*, 546(7656):101–106, 2017.
- B. K. Hulse et al. A connectome of the *Drosophila* central complex reveals network motifs suitable for flexible navigation and context-dependent action selection. *eLife*, 10:e66039, 2021.
- S. S. Kim, H. Rouault, S. Druckmann, and V. Jayaraman. Ring attractor dynamics in the *Drosophila* central brain. *Science*, 356(6340):849–853, 2017.
- J. D. Seelig and V. Jayaraman. Neural dynamics for landmark orientation and angular path integration. *Nature*, 521(7551):186–191, 2015.
- P. Vafidis, D. Oswald, T. D’Albis, and R. Kempter. Learning accurate path integration in ring attractor models of the head direction system. *eLife*, 11:e69841, 2022.